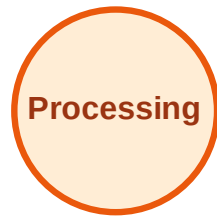


BFS Traversal

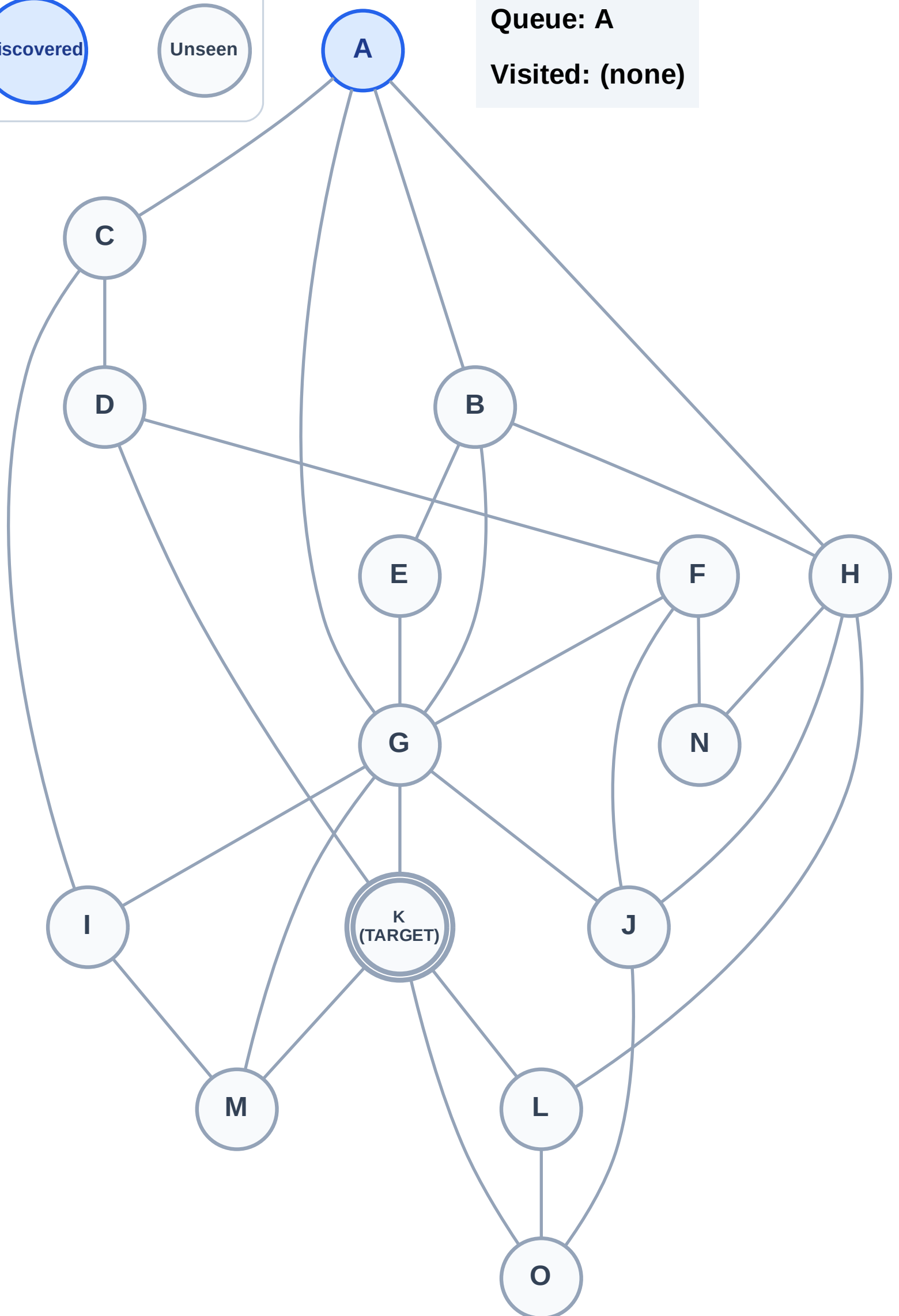
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

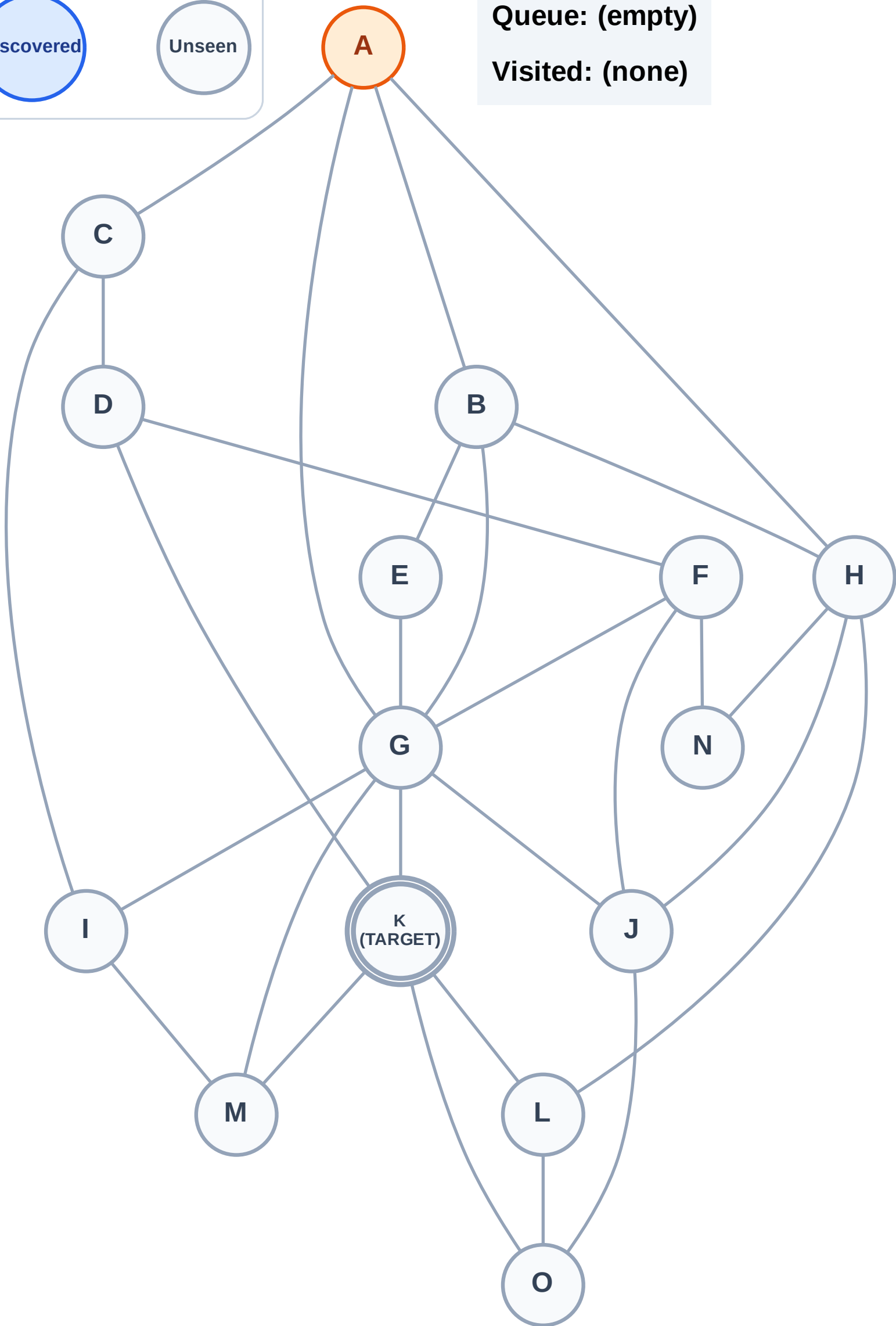
Complete

Processing

Discovered

Unseen

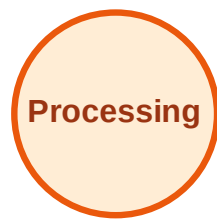
Queue: (empty)
Visited: (none)



BFS Traversal

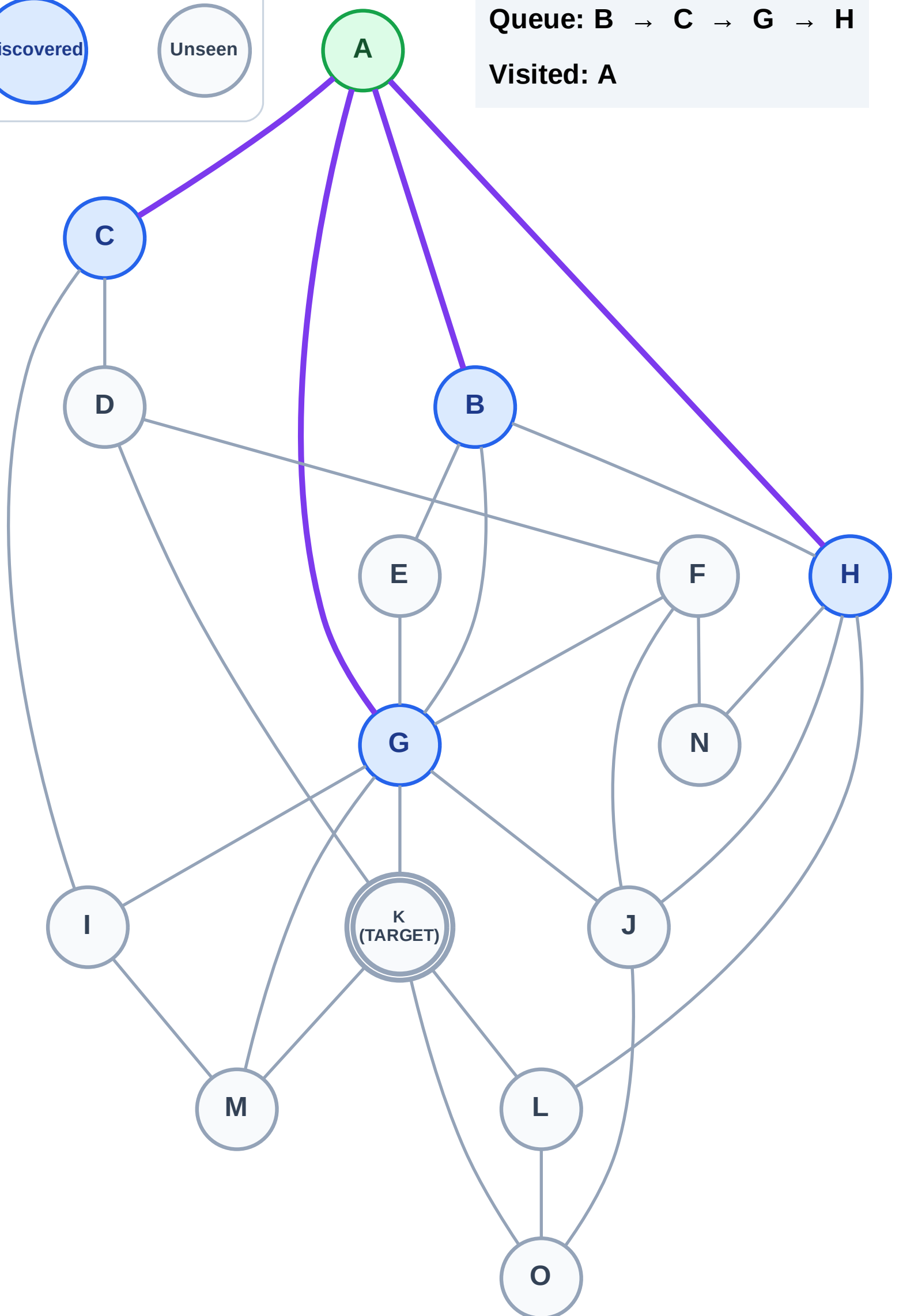
Step 3: Discover B, C, G, H from A

Legend



Queue: B → C → G → H

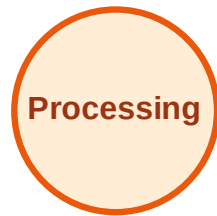
Visited: A



BFS Traversal

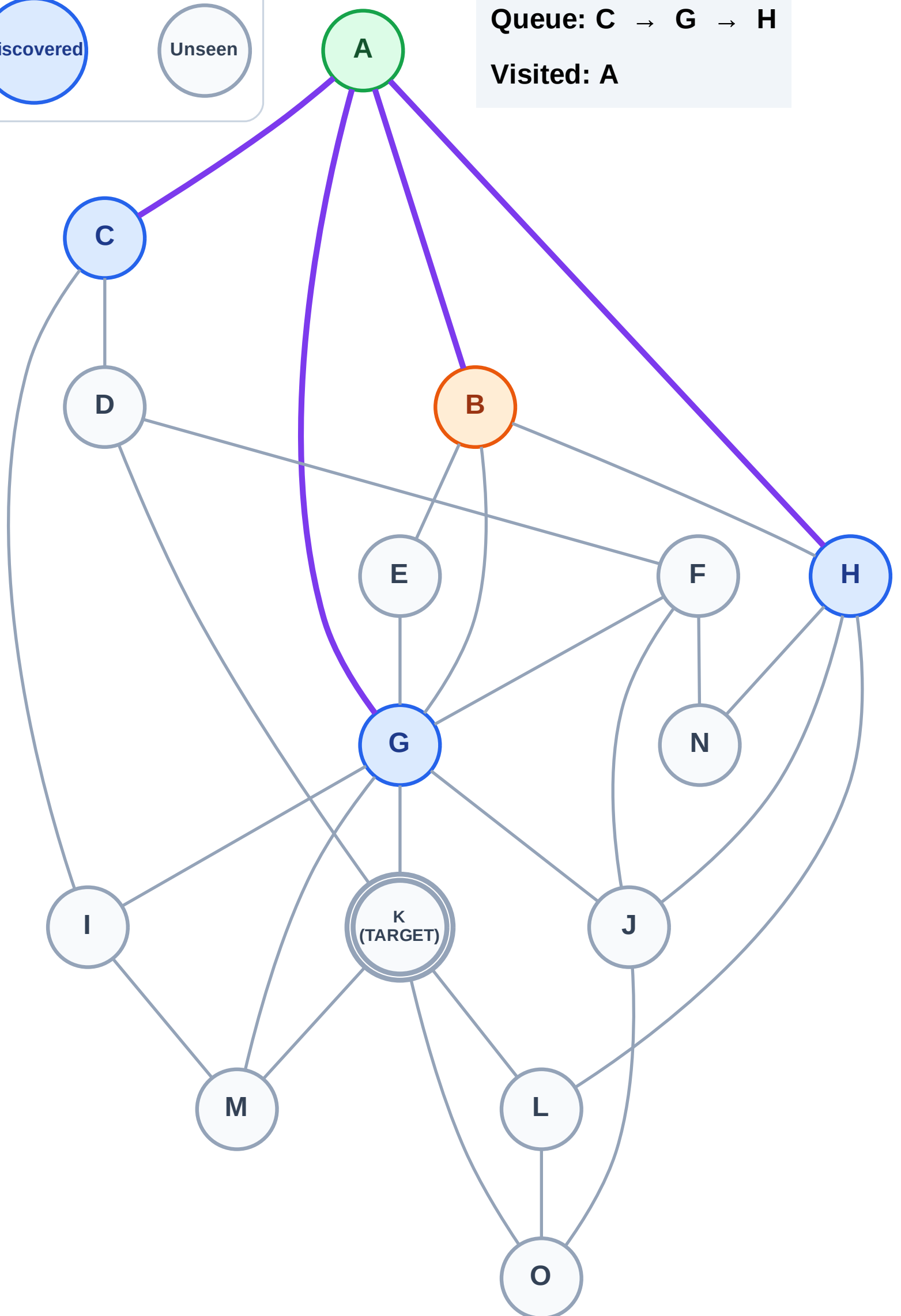
Step 4: Process node B

Legend



Queue: C → G → H

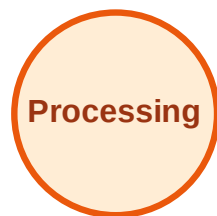
Visited: A



BFS Traversal

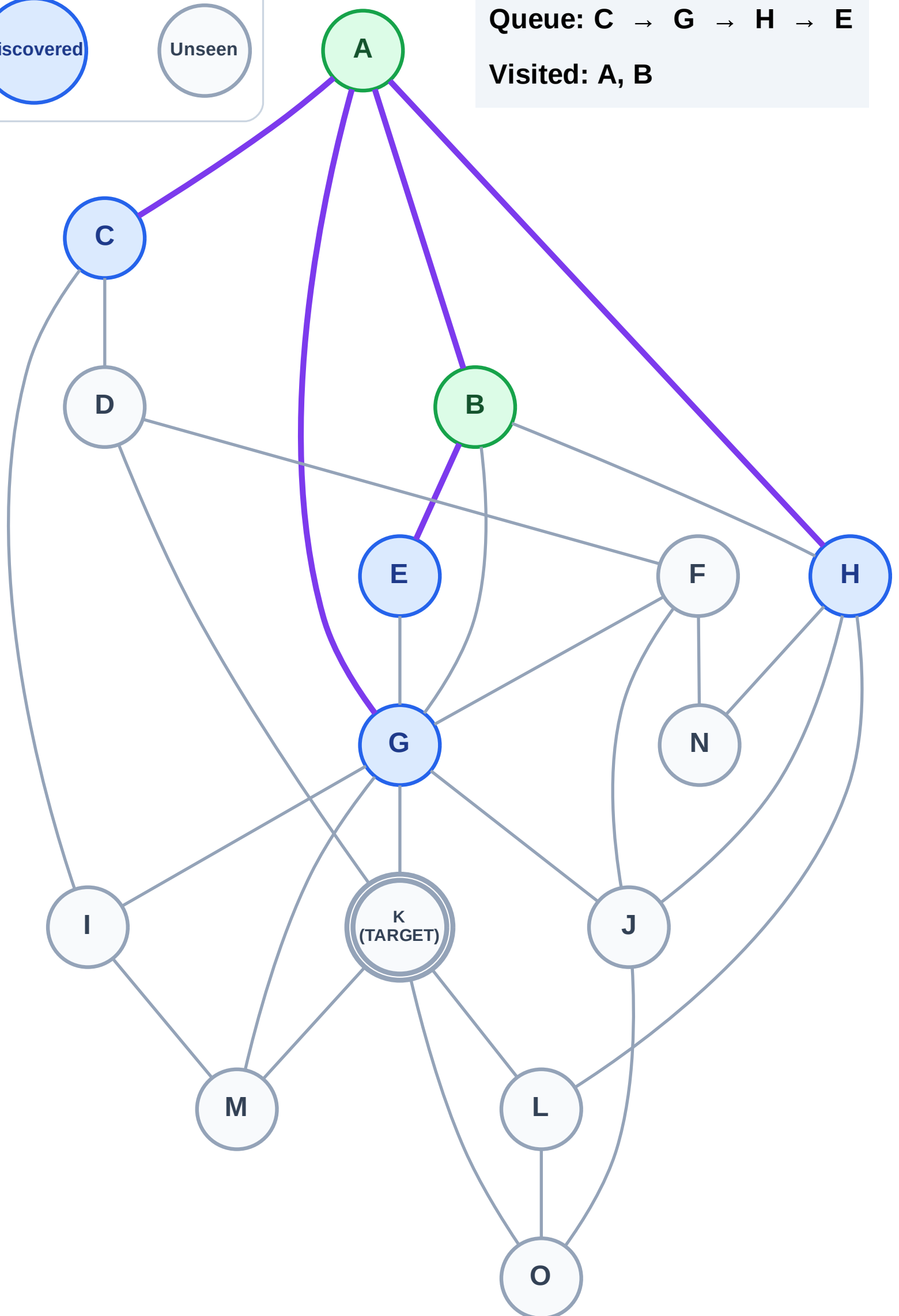
Step 5: Discover E from B

Legend



Queue: C → G → H → E

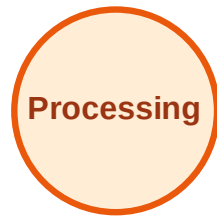
Visited: A, B



BFS Traversal

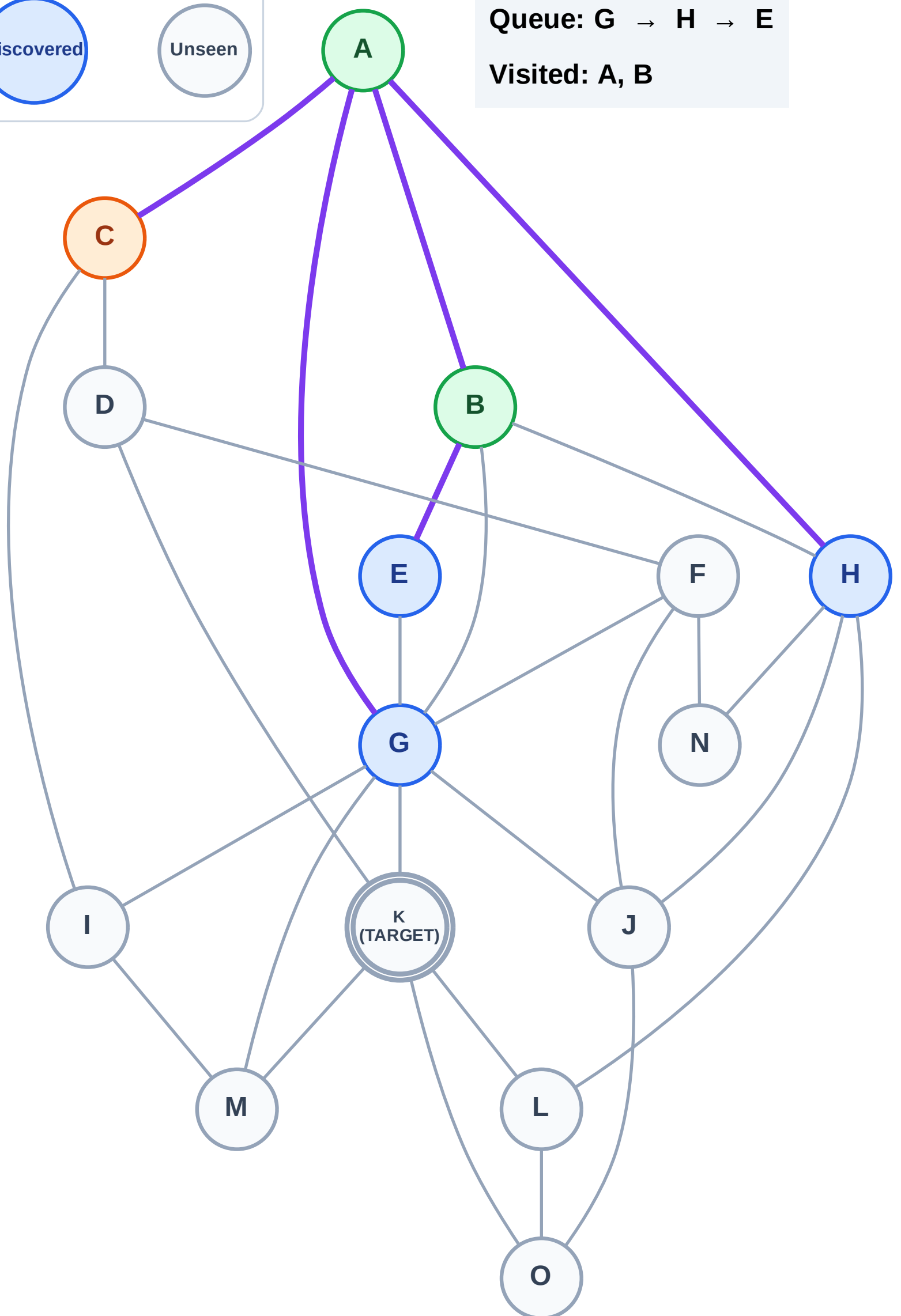
Step 6: Process node C

Legend



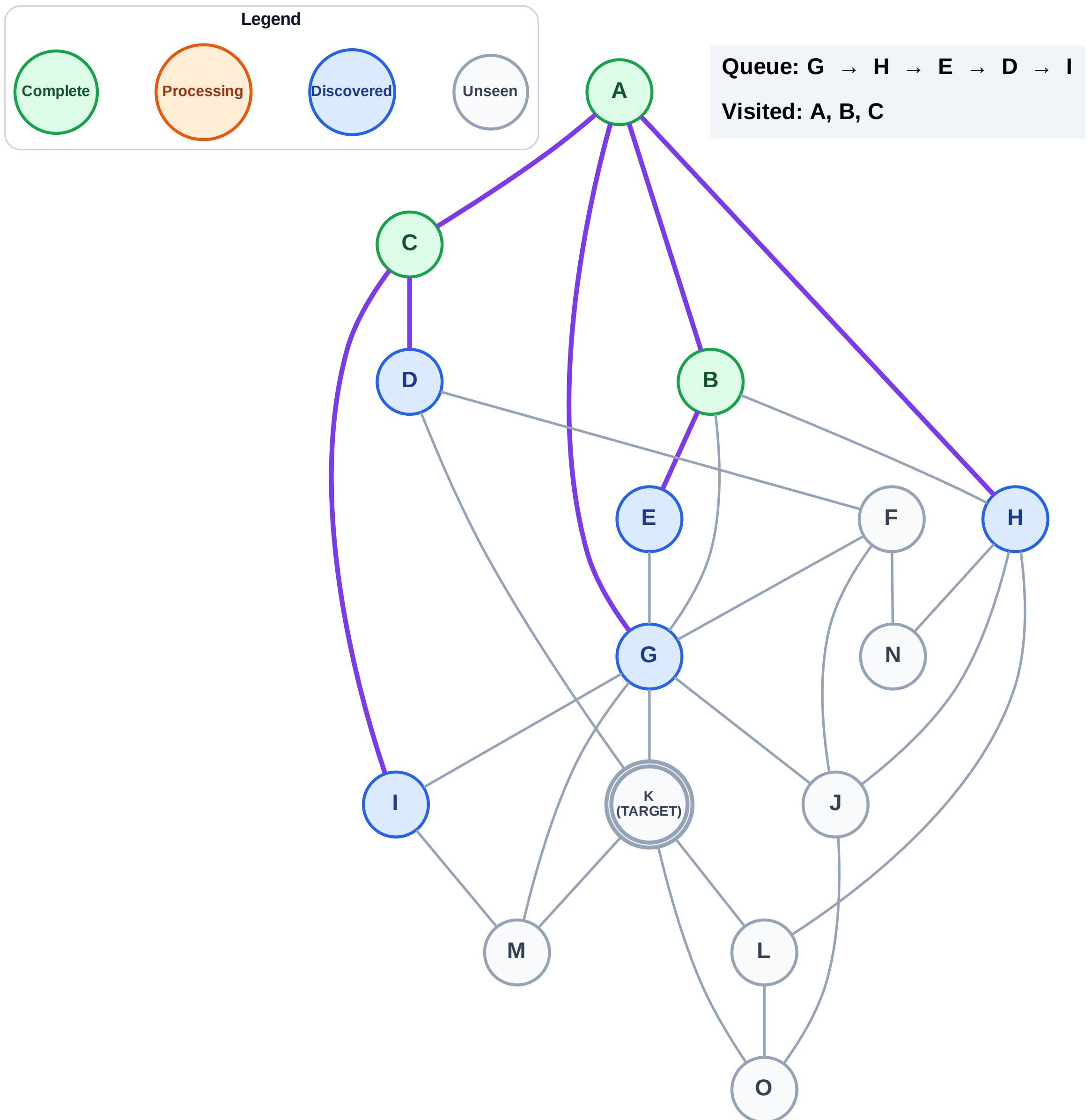
Queue: G → H → E

Visited: A, B



Step 7: Discover D, I from C

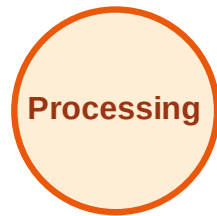
Step 7: Discover D, I from C



BFS Traversal

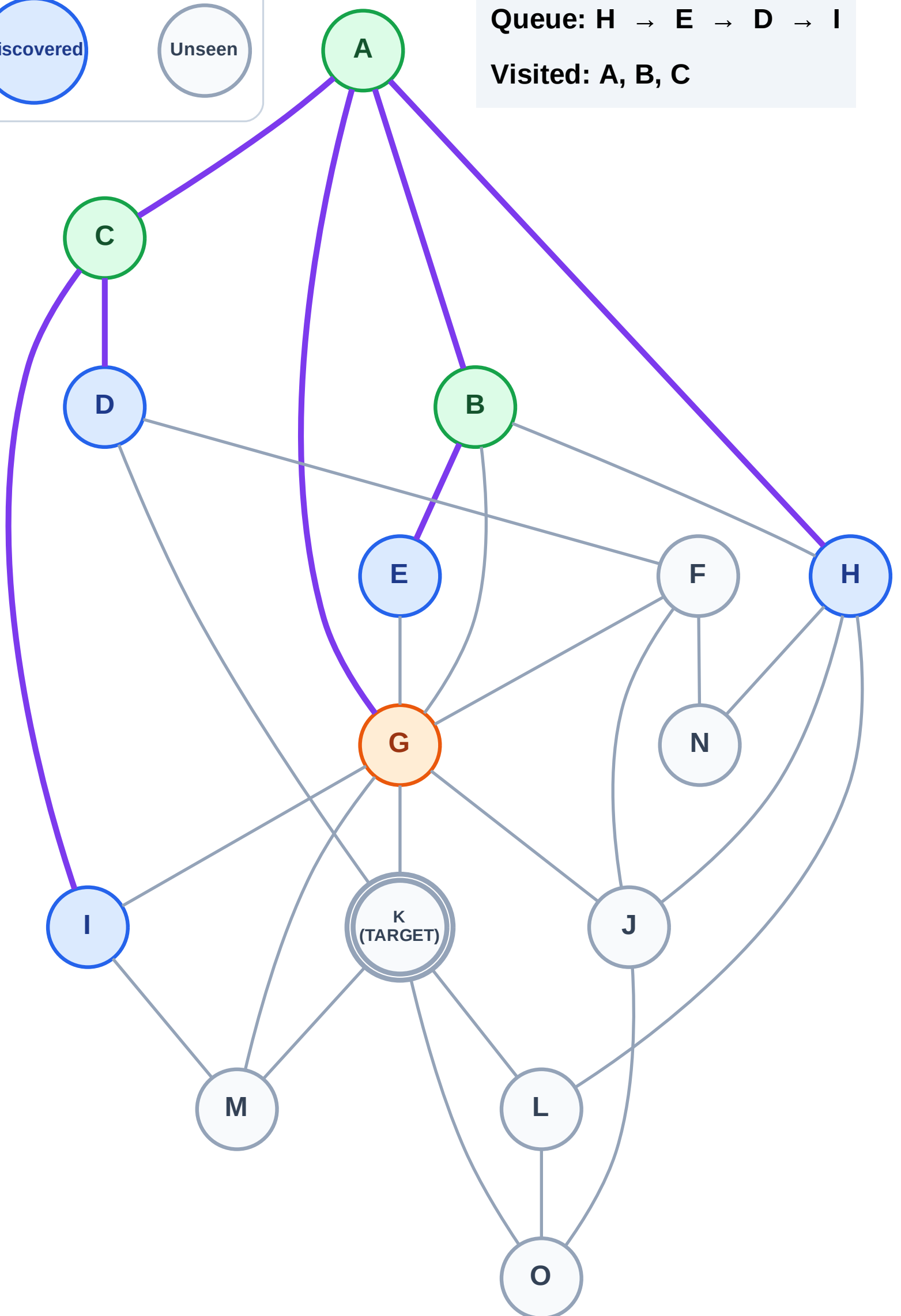
Step 8: Process node G

Legend



Queue: H → E → D → I

Visited: A, B, C



BFS Traversal

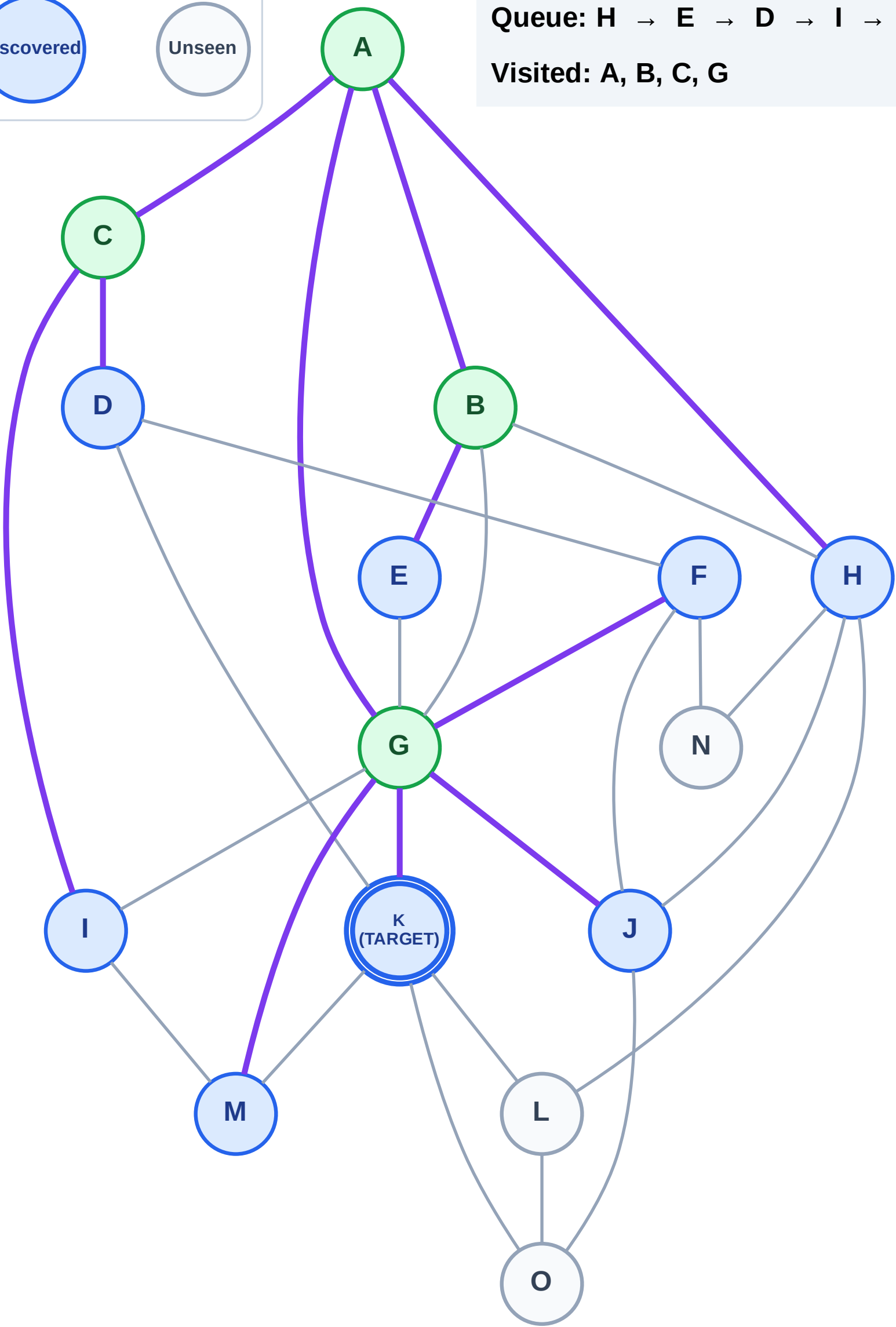
Step 9: Discover F, J, K, M from G

Legend



Queue: H → E → D → I → F → J → K → M

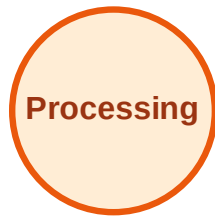
Visited: A, B, C, G



BFS Traversal

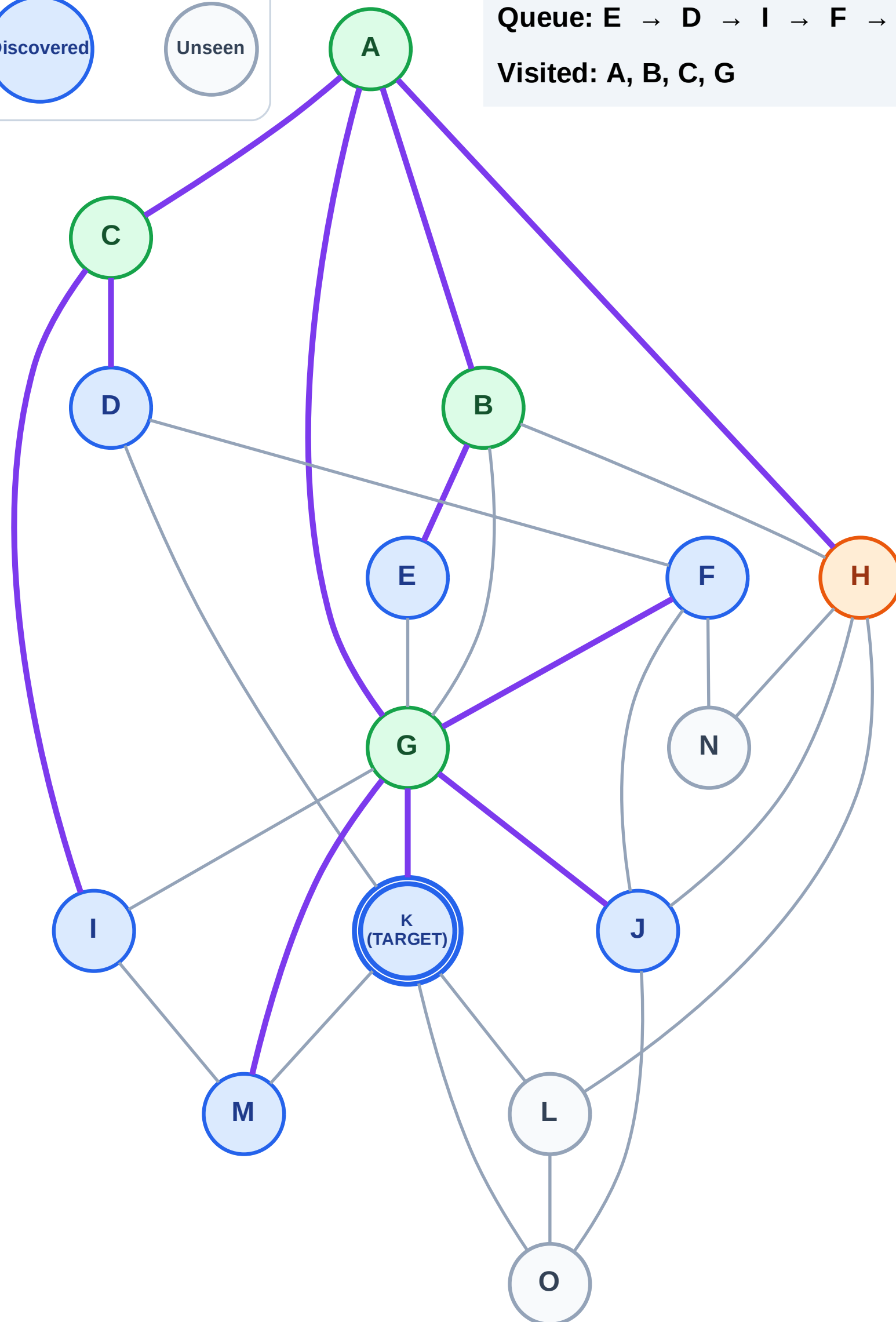
Step 10: Process node H

Legend



Queue: E → D → I → F → J → K → M

Visited: A, B, C, G



BFS Traversal

Step 11: Discover L, N from H

Legend

Complete

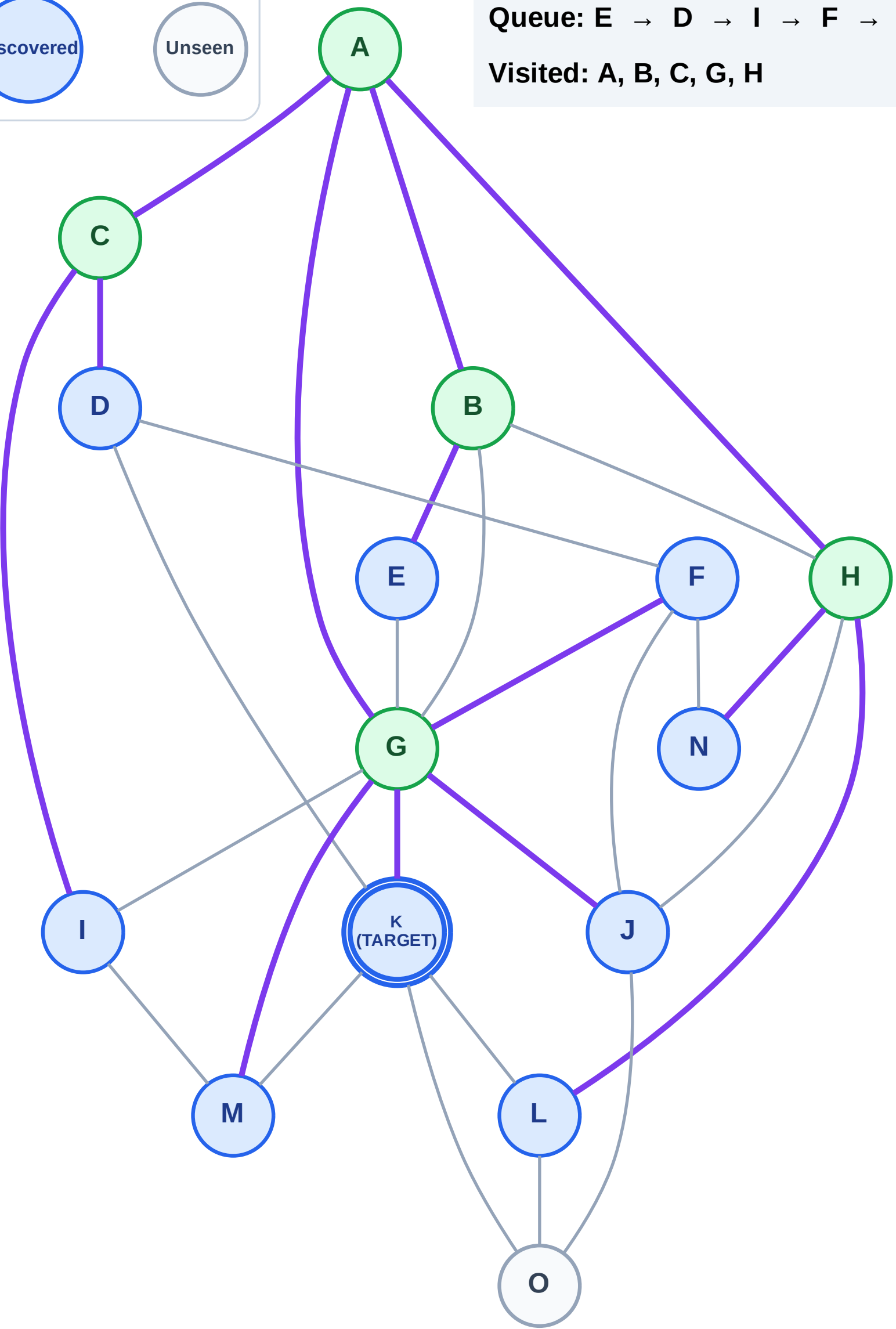
Processing

Discovered

Unseen

Queue: E → D → I → F → J → K → M → L → N

Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node E

Legend

Complete

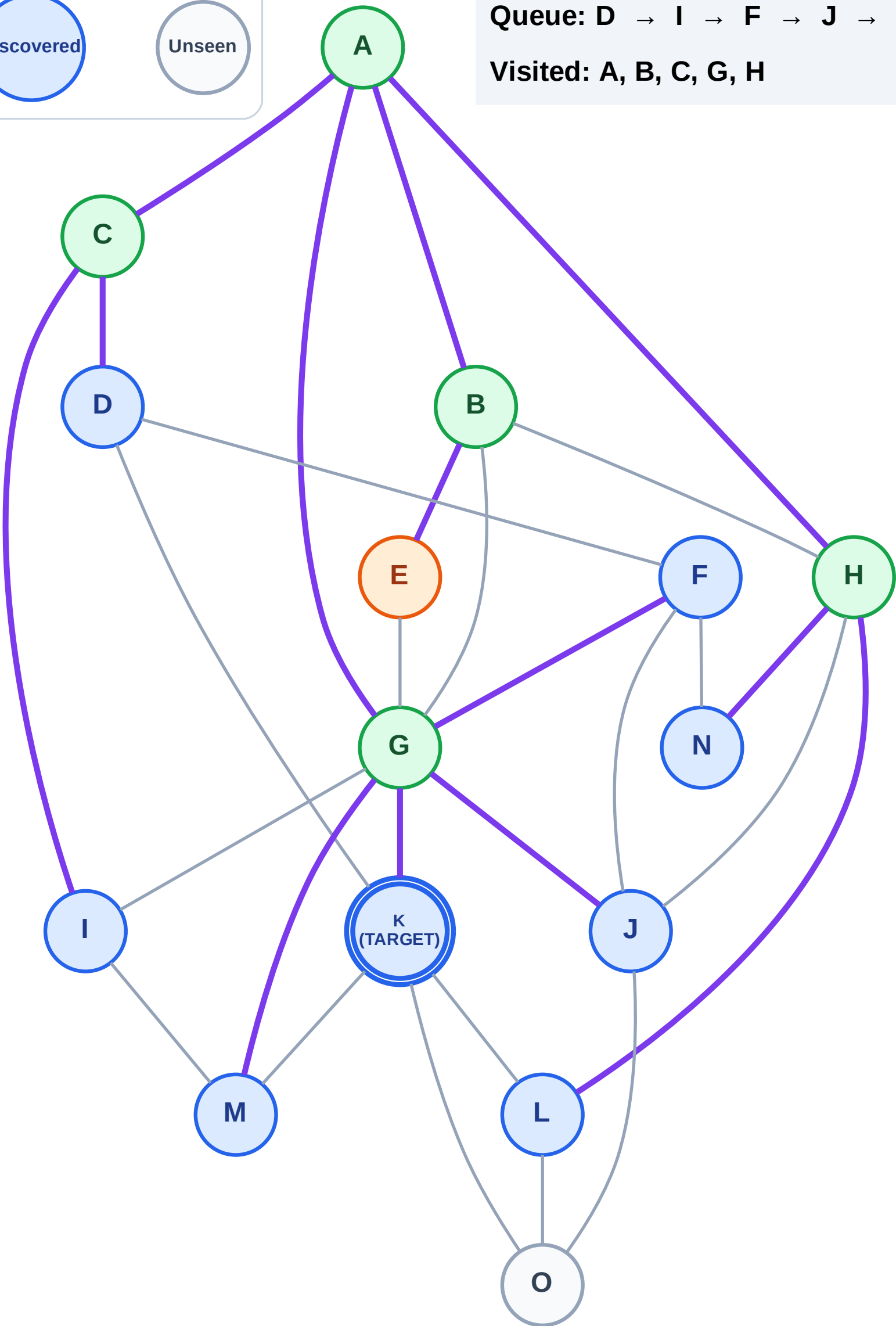
Processing

Discovered

Unseen

Queue: D → I → F → J → K → M → L → N

Visited: A, B, C, G, H



BFS Traversal

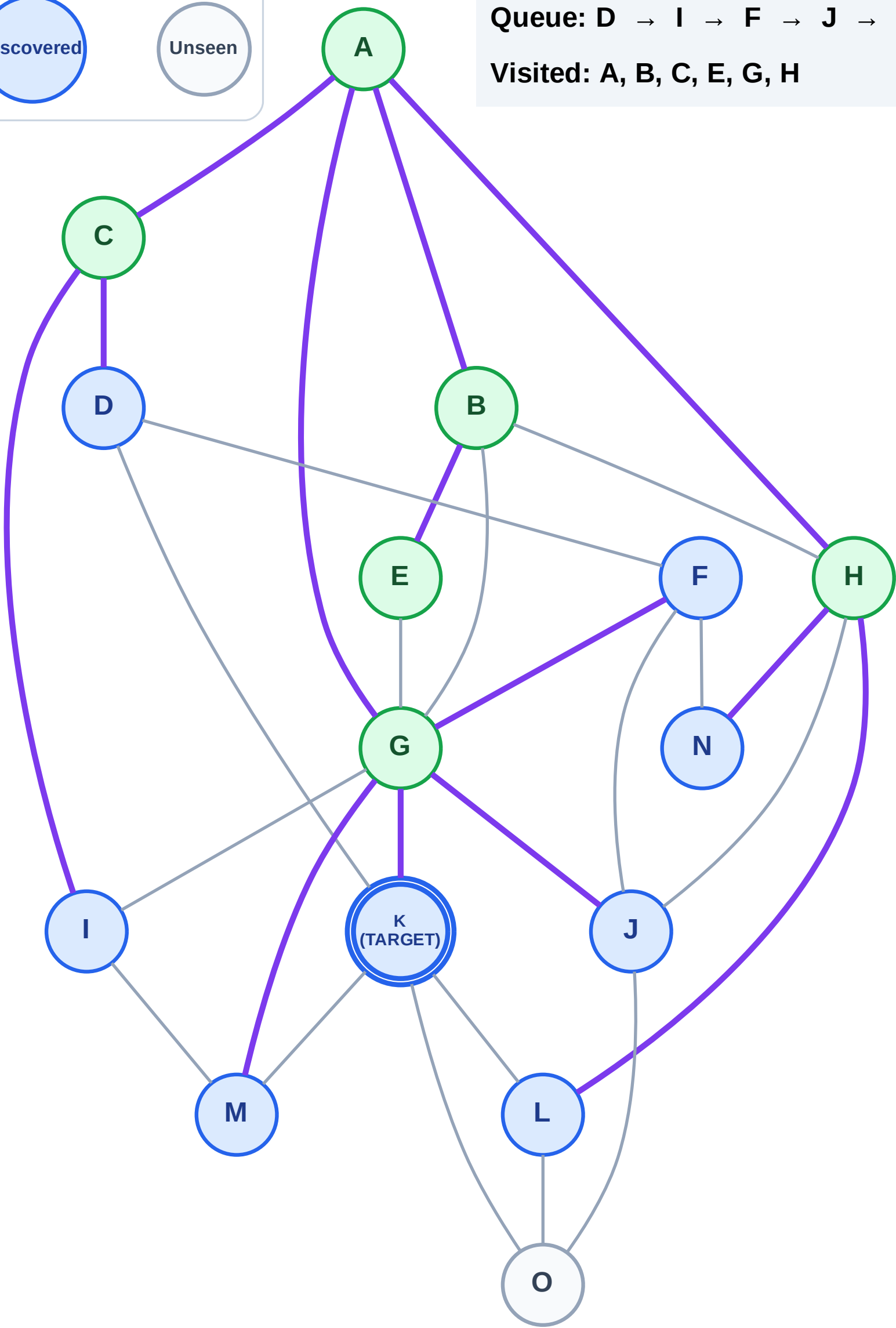
Step 13: No new neighbors from E

Legend



Queue: D → I → F → J → K → M → L → N

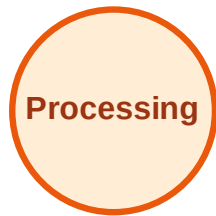
Visited: A, B, C, E, G, H



BFS Traversal

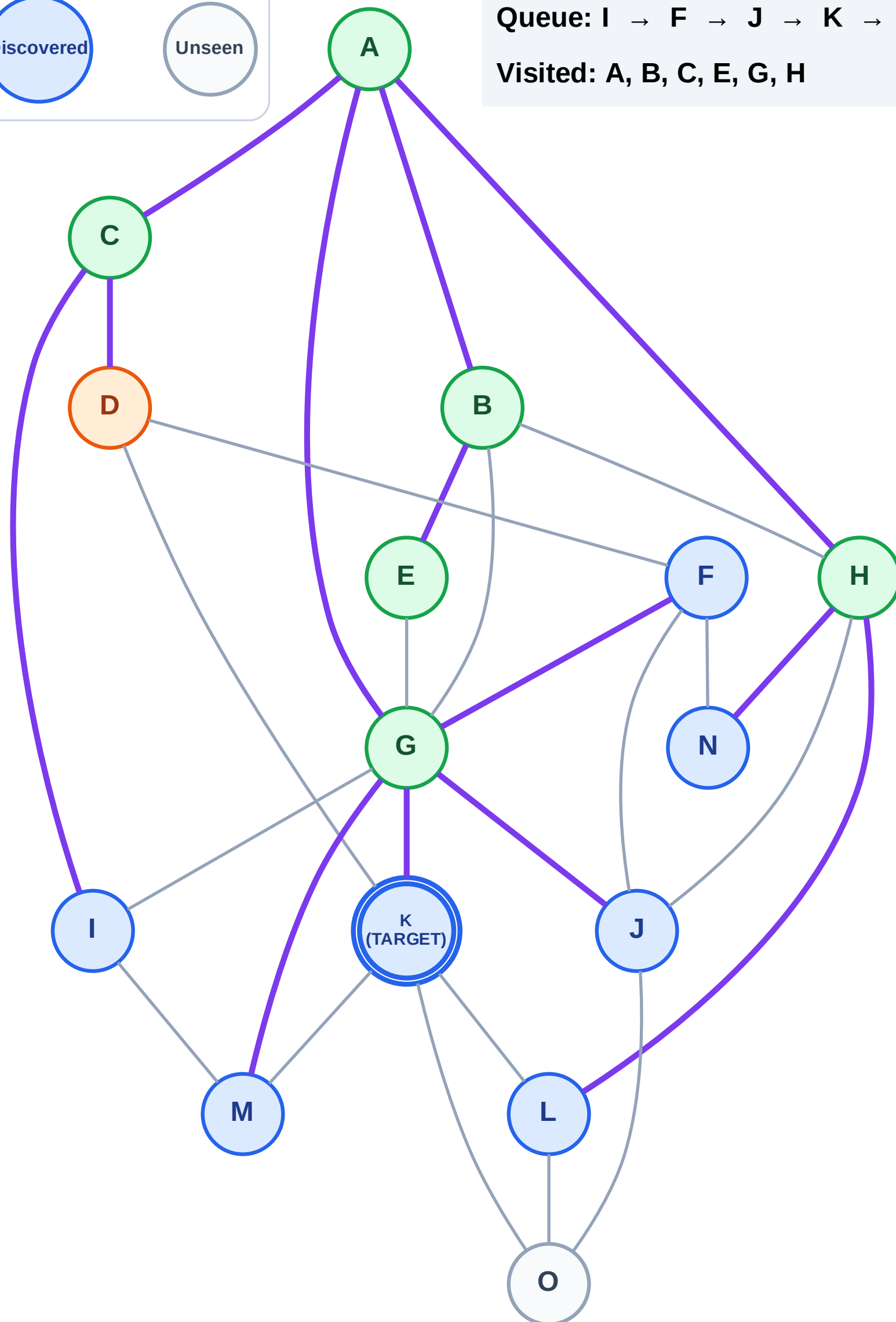
Step 14: Process node D

Legend



Queue: I → F → J → K → M → L → N

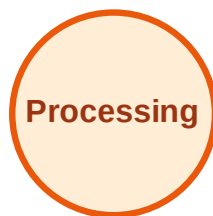
Visited: A, B, C, E, G, H



BFS Traversal

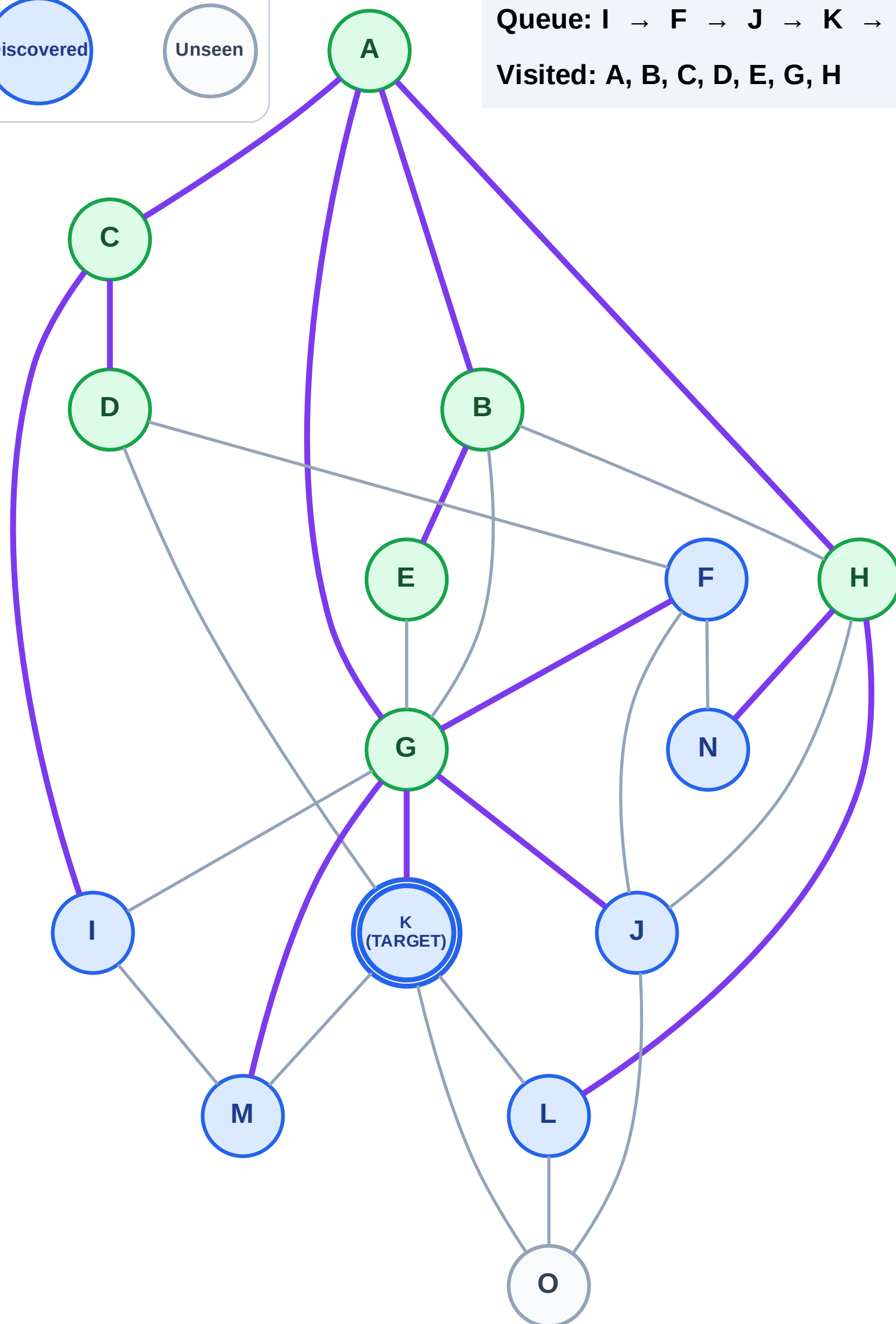
Step 15: No new neighbors from D

Legend



Queue: I → F → J → K → M → L → N

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

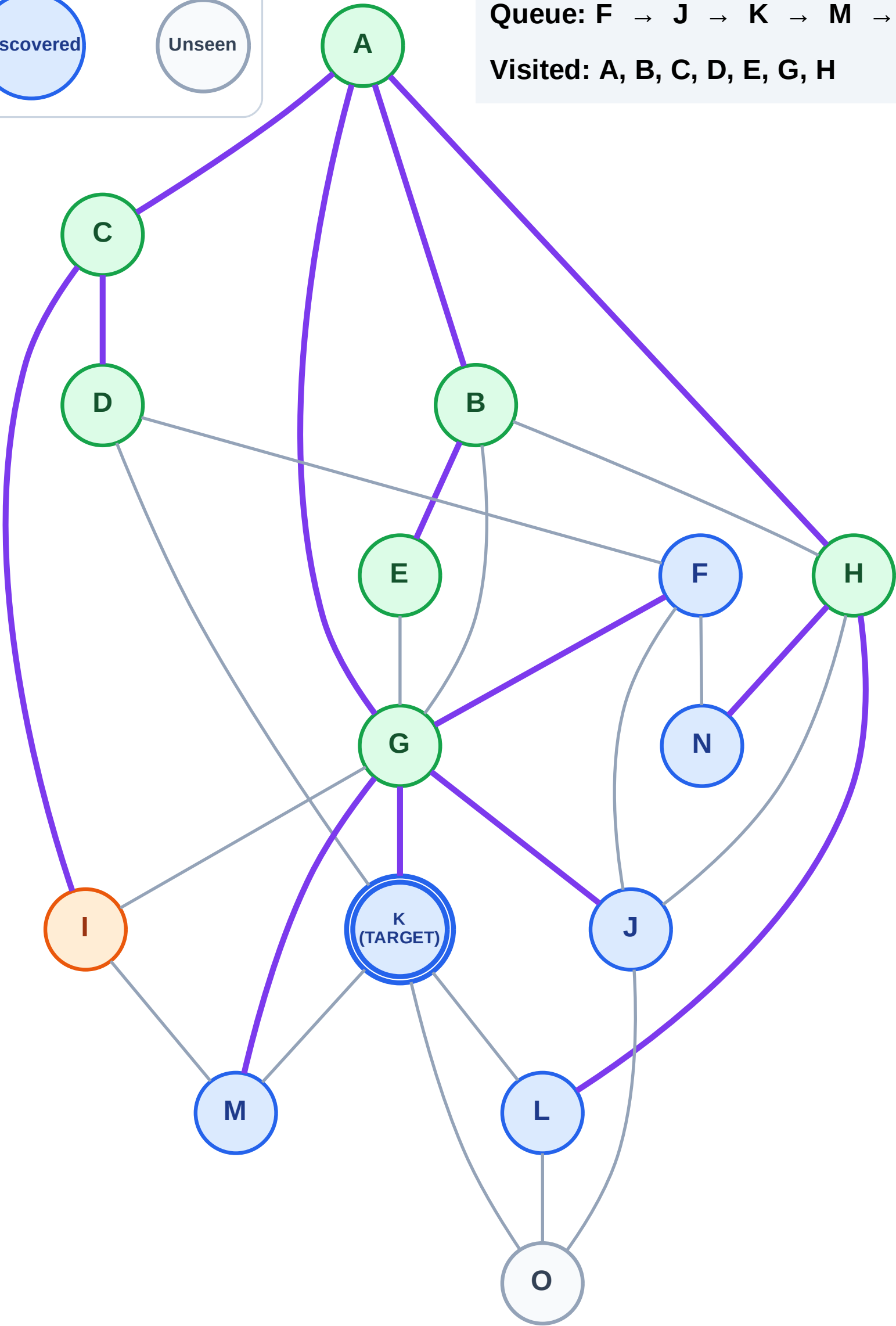
Processing

Discovered

Unseen

Queue: F → J → K → M → L → N

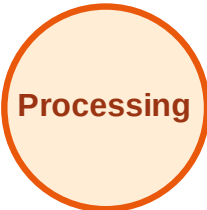
Visited: A, B, C, D, E, G, H



BFS Traversal

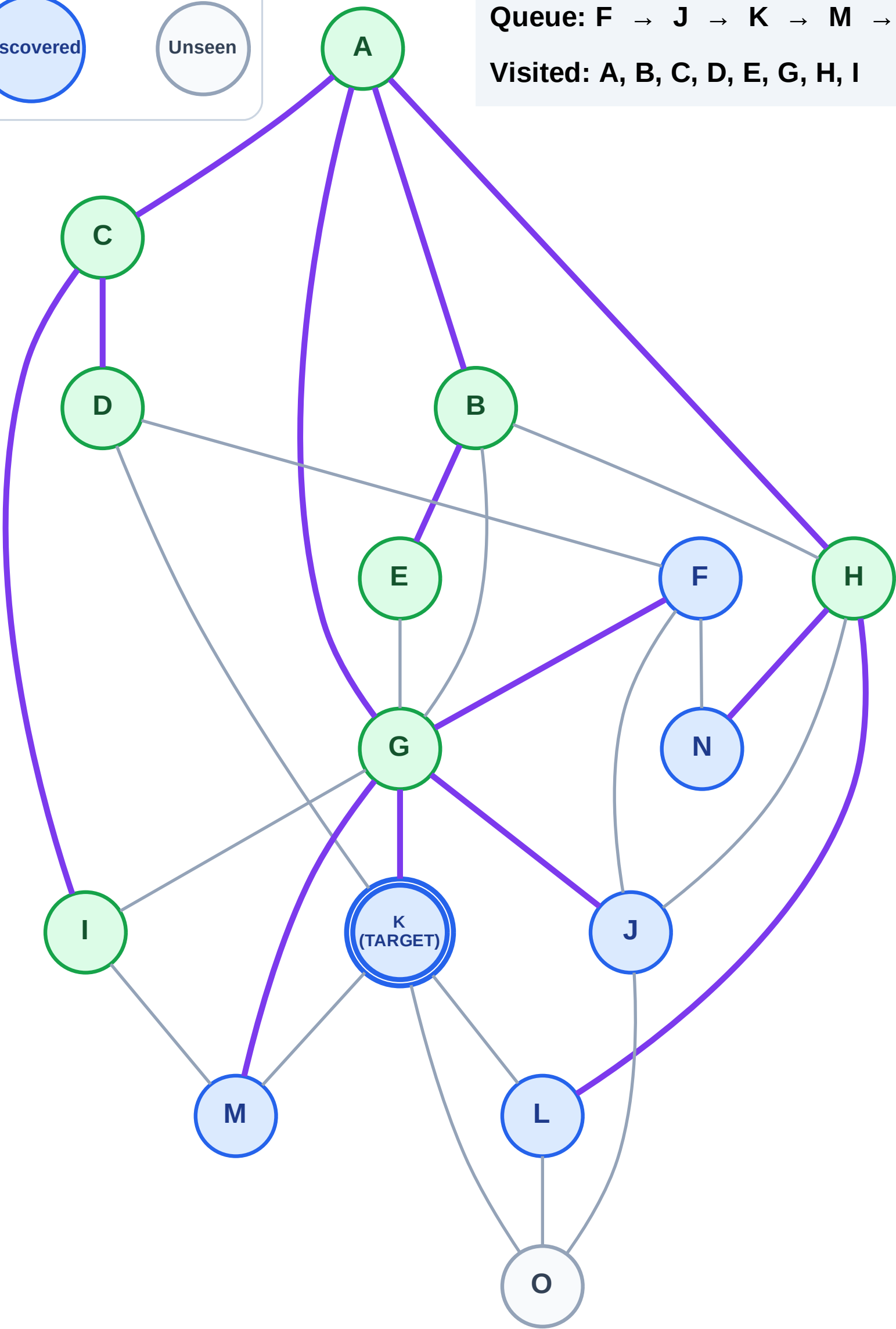
Step 17: No new neighbors from I

Legend



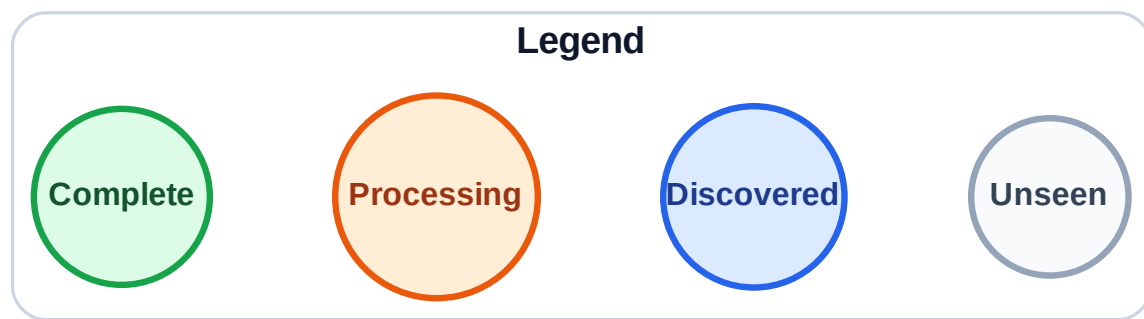
Queue: F → J → K → M → L → N

Visited: A, B, C, D, E, G, H, I



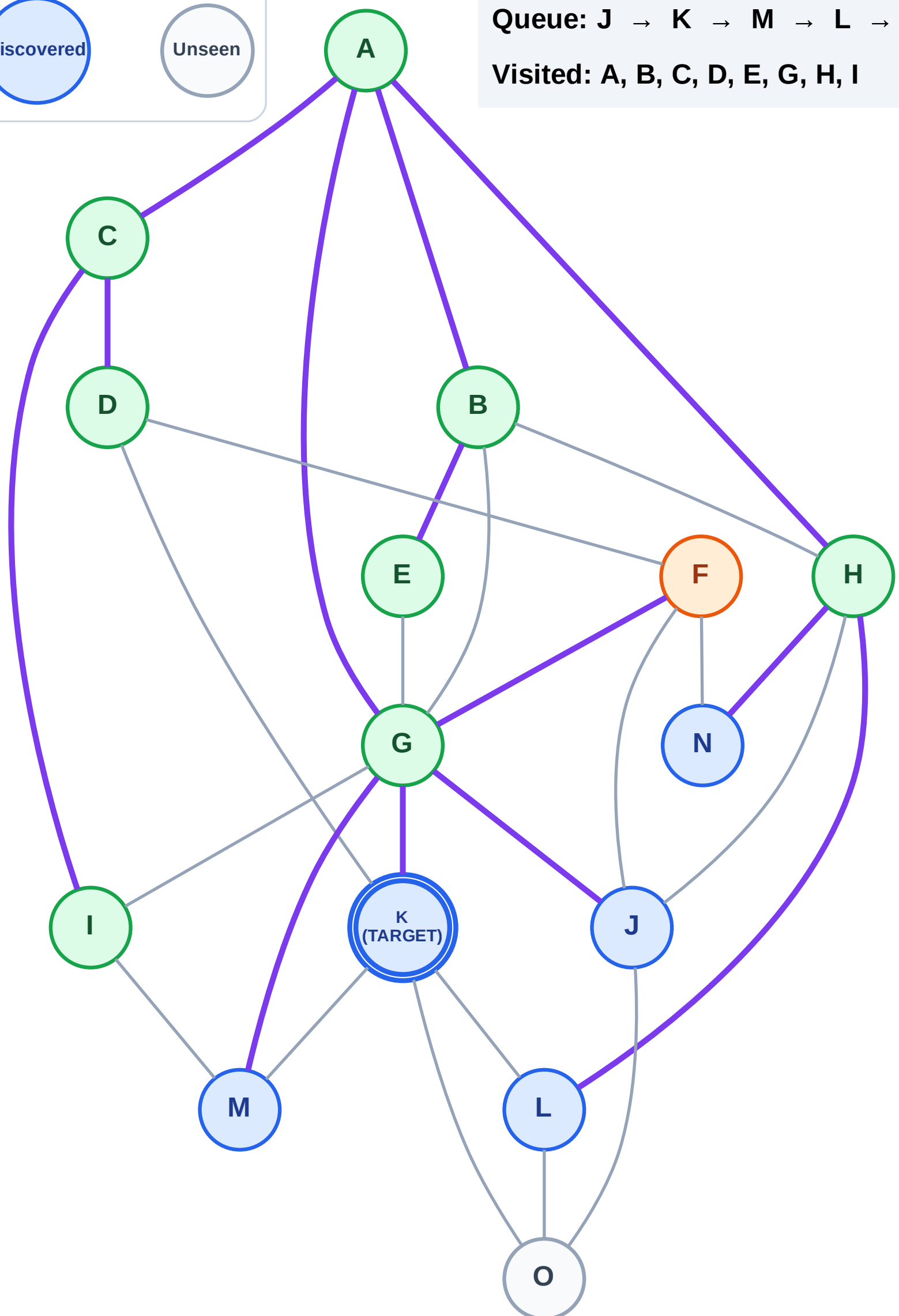
BFS Traversal

Step 18: Process node F



Queue: J → K → M → L → N

Visited: A, B, C, D, E, G, H, I



BFS Traversal

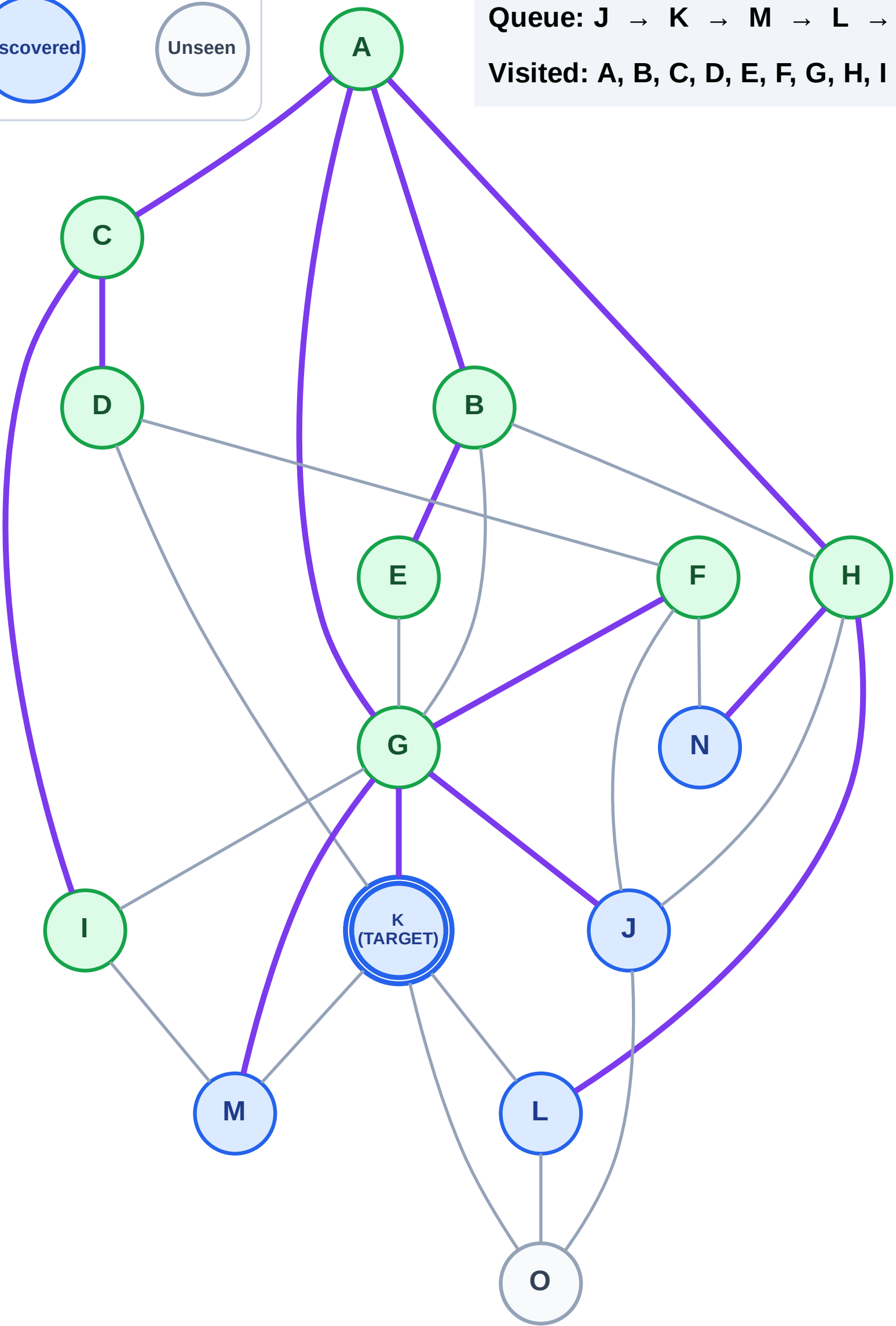
Step 19: No new neighbors from F

Legend



Queue: J → K → M → L → N

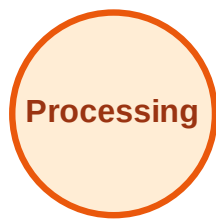
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

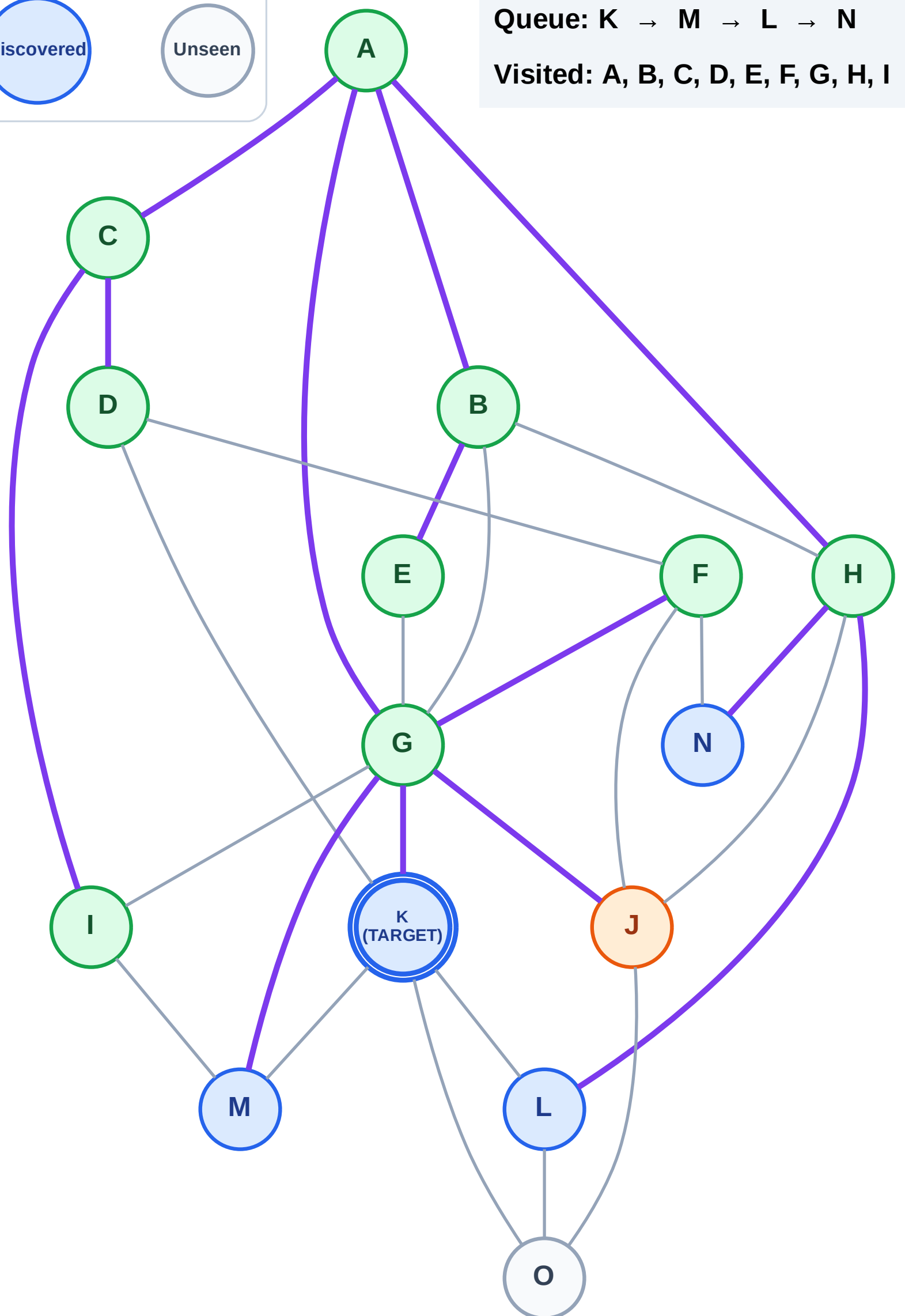
Step 20: Process node J

Legend



Queue: K → M → L → N

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

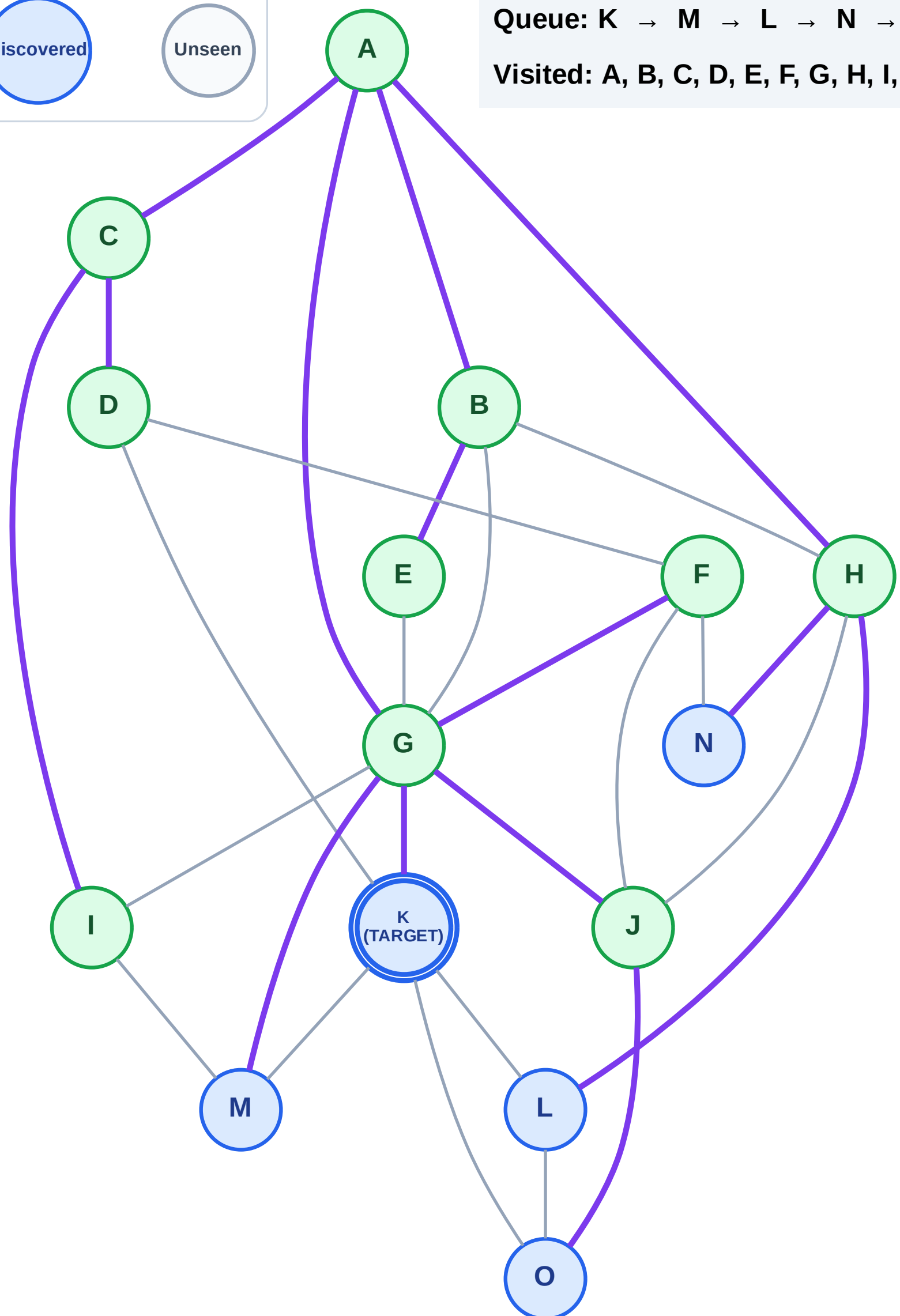
Step 21: Discover O from J

Legend



Queue: K → M → L → N → O

Visited: A, B, C, D, E, F, G, H, I, J



BFS Traversal

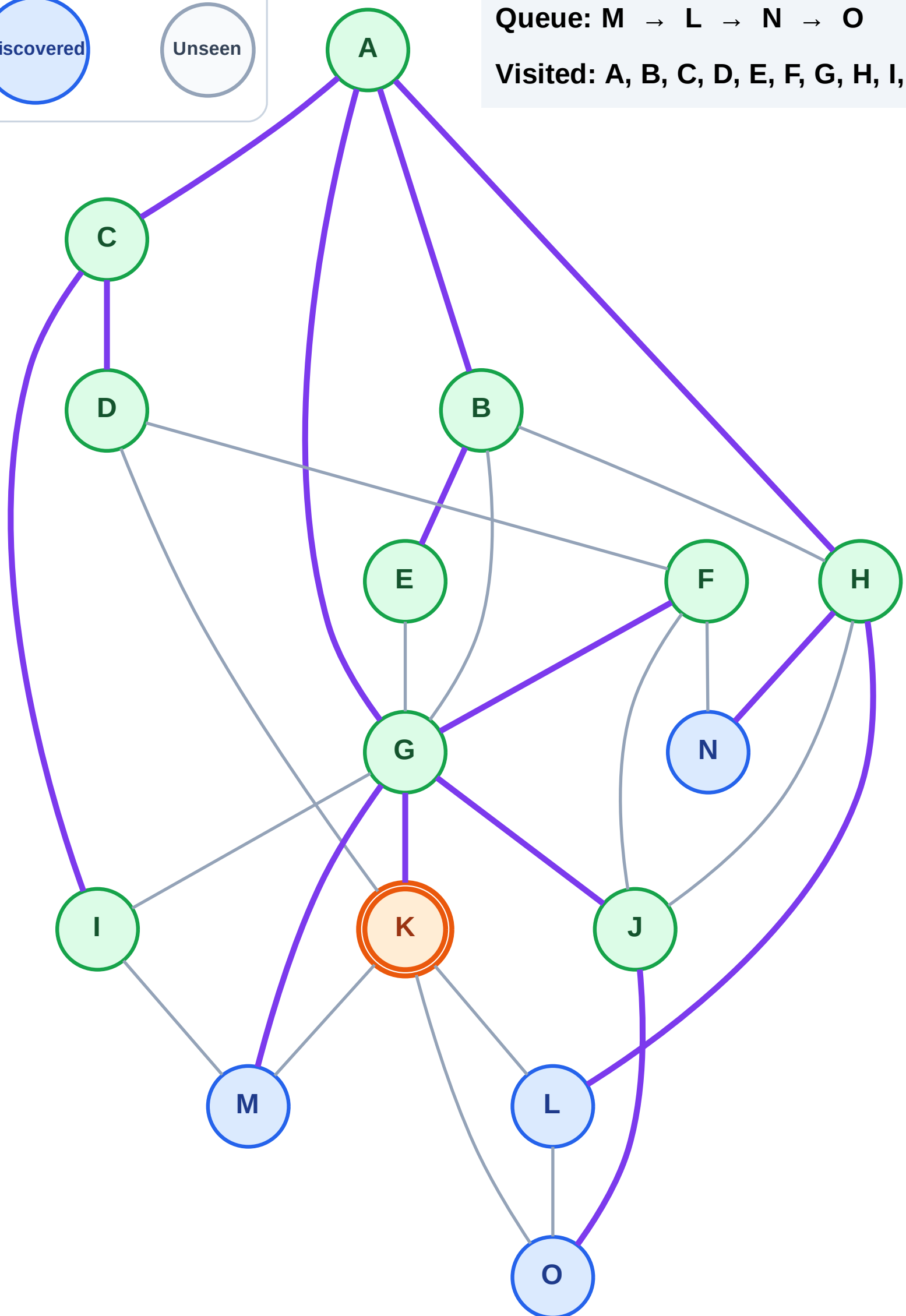
Step 22: Process node K

Legend



Queue: M → L → N → O

Visited: A, B, C, D, E, F, G, H, I, J



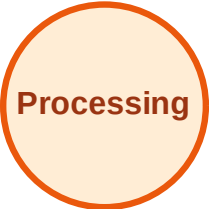
BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend



Complete



Processing



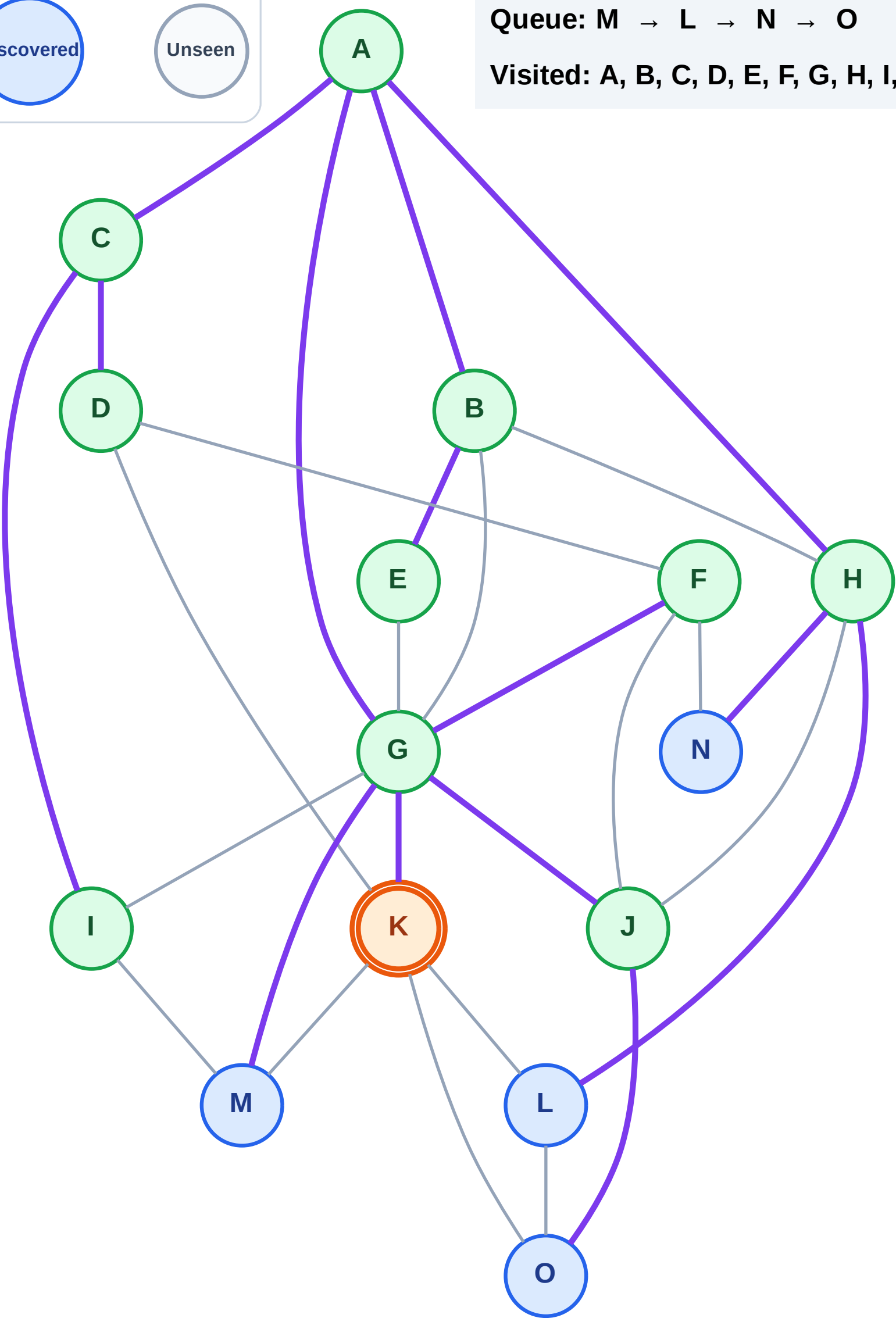
Discovered



Unseen

Queue: M → L → N → O

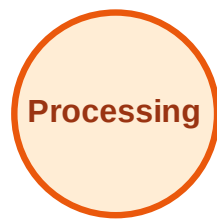
Visited: A, B, C, D, E, F, G, H, I, J, K



DFS Traversal

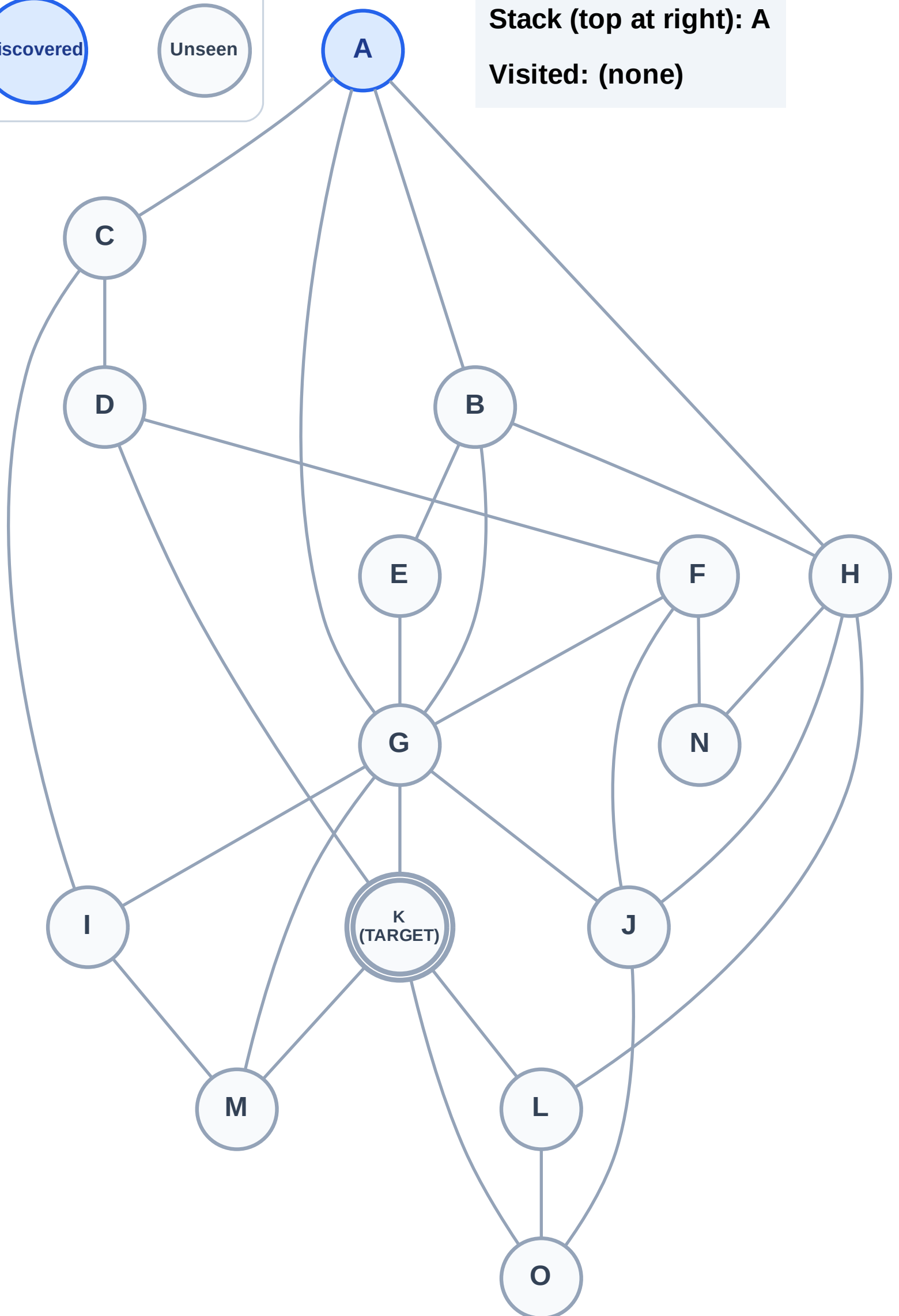
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

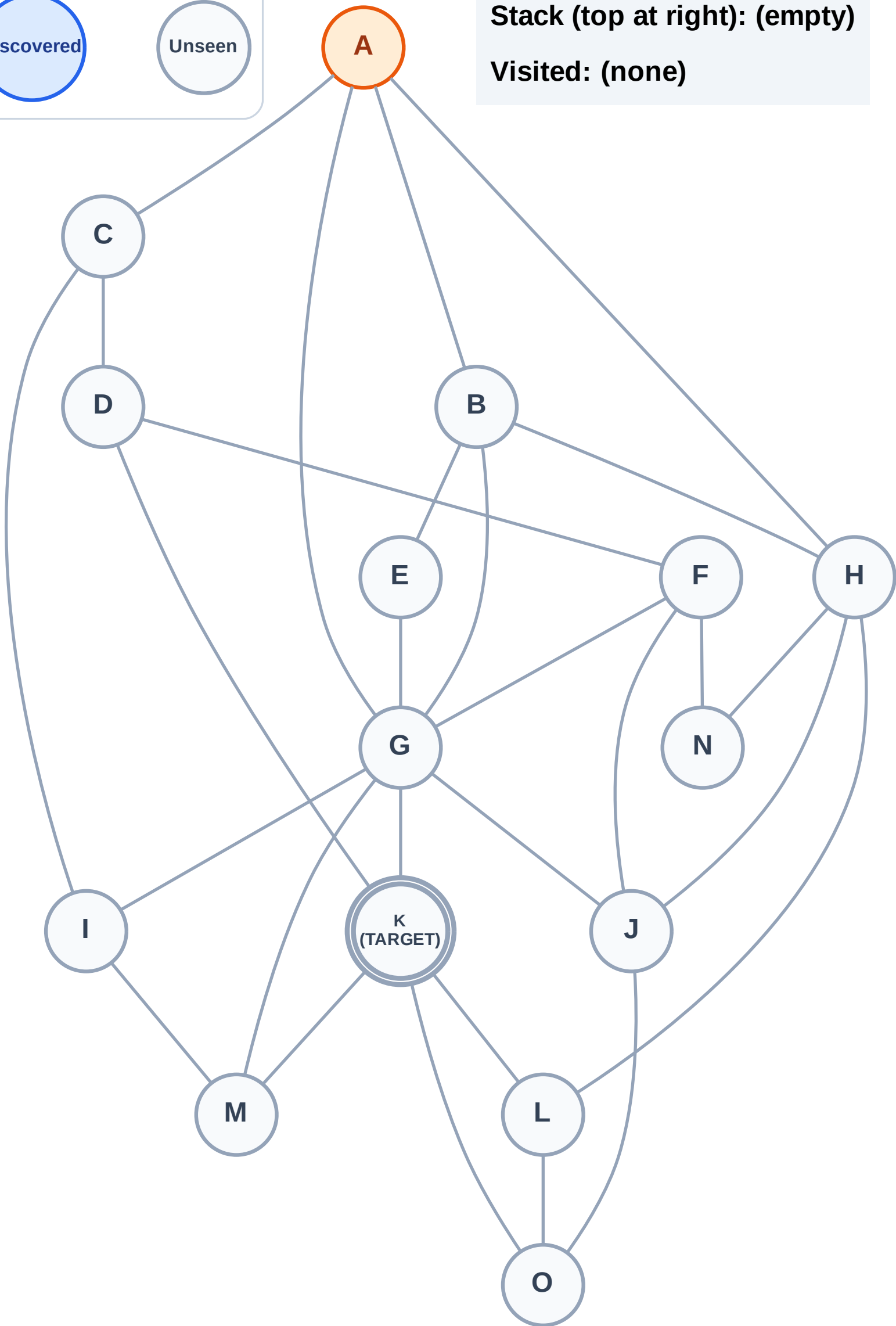
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, G, C, B from A

Legend

Complete

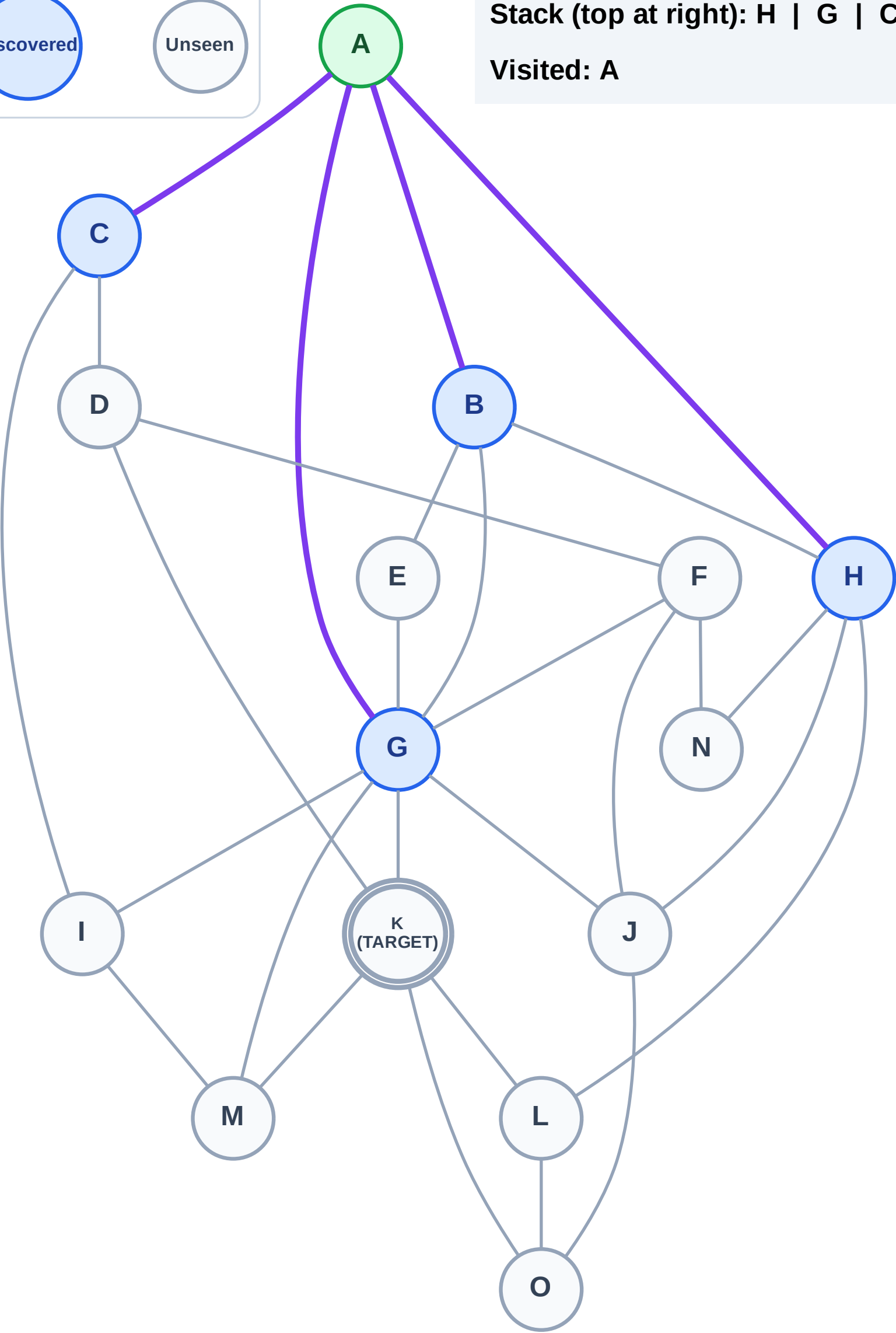
Processing

Discovered

Unseen

Stack (top at right): H | G | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

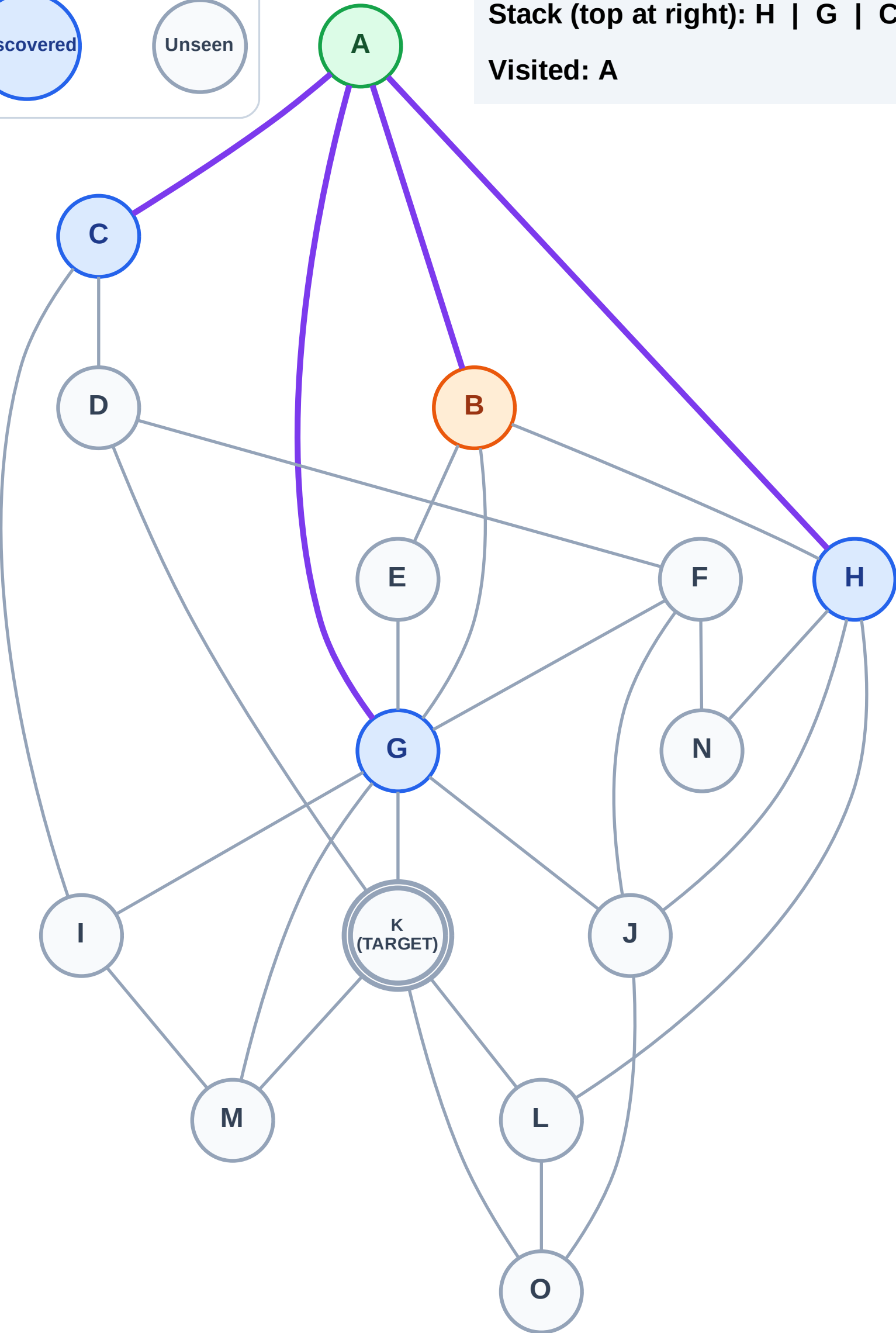
Processing

Discovered

Unseen

Stack (top at right): H | G | C

Visited: A



DFS Traversal

Step 5: Discover E from B

Legend

Complete

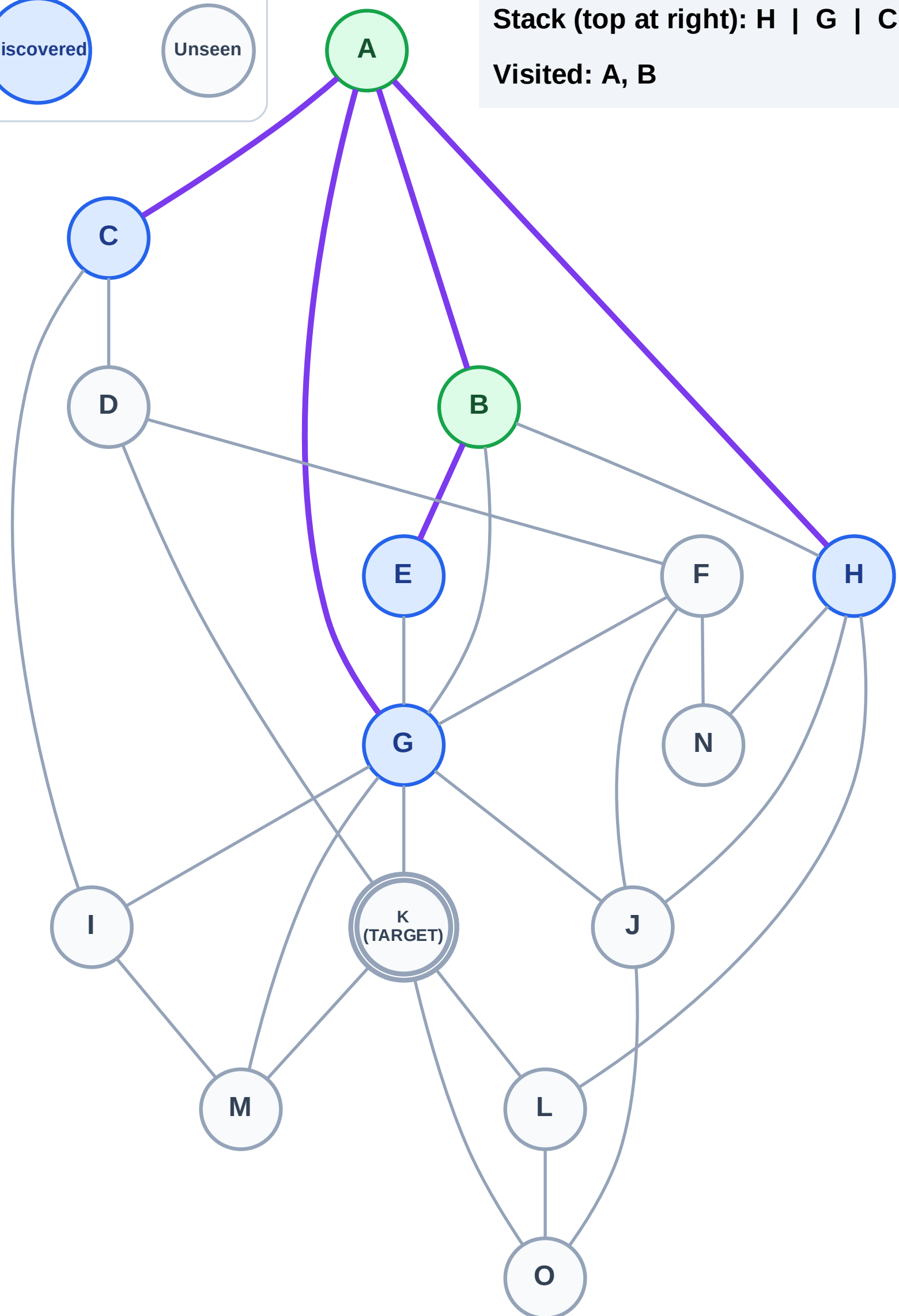
Processing

Discovered

Unseen

Stack (top at right): H | G | C | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

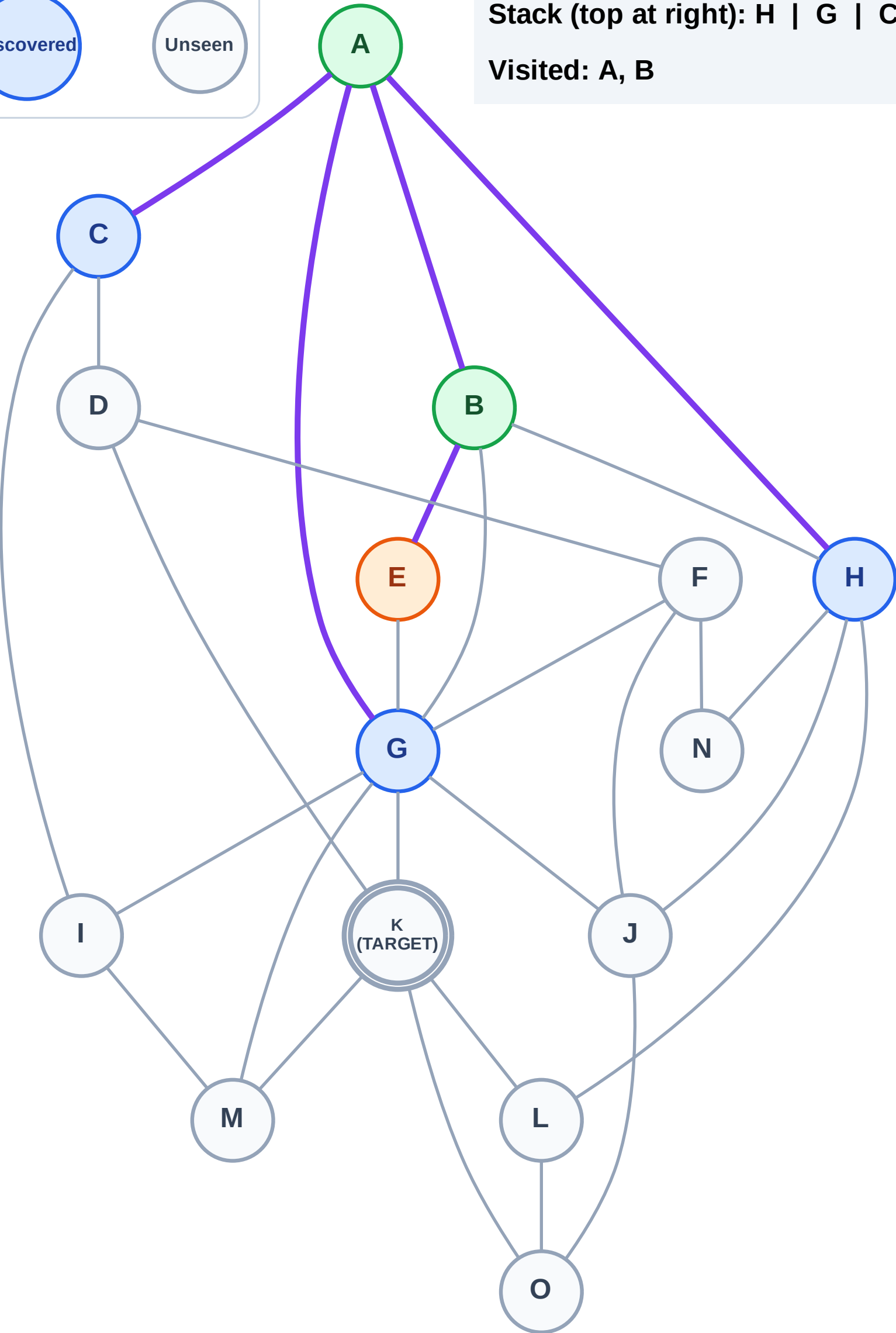
Processing

Discovered

Unseen

Stack (top at right): H | G | C

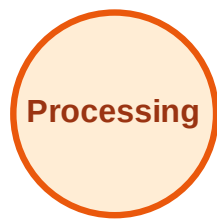
Visited: A, B



DFS Traversal

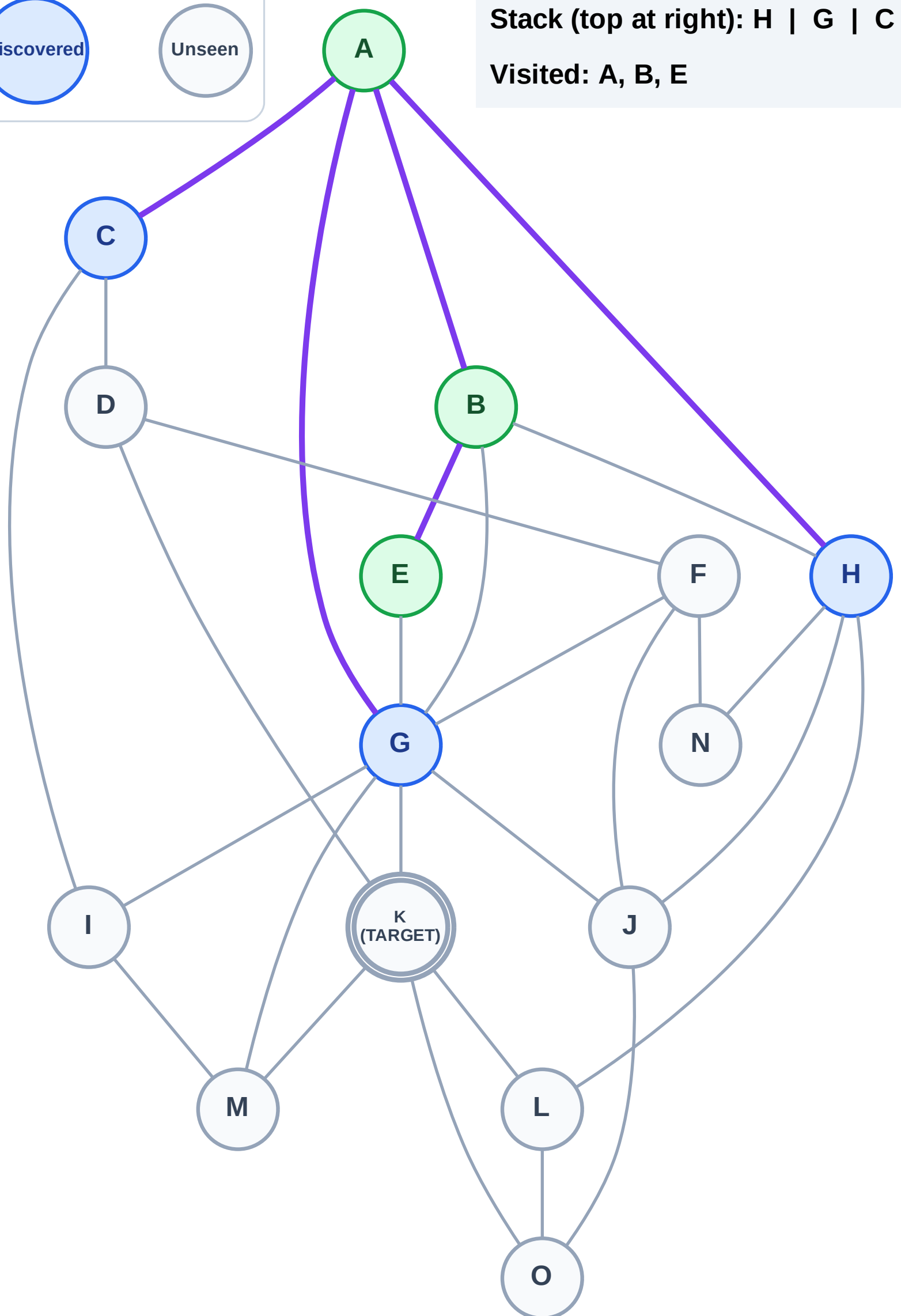
Step 7: No new neighbors from E

Legend



Stack (top at right): H | G | C

Visited: A, B, E

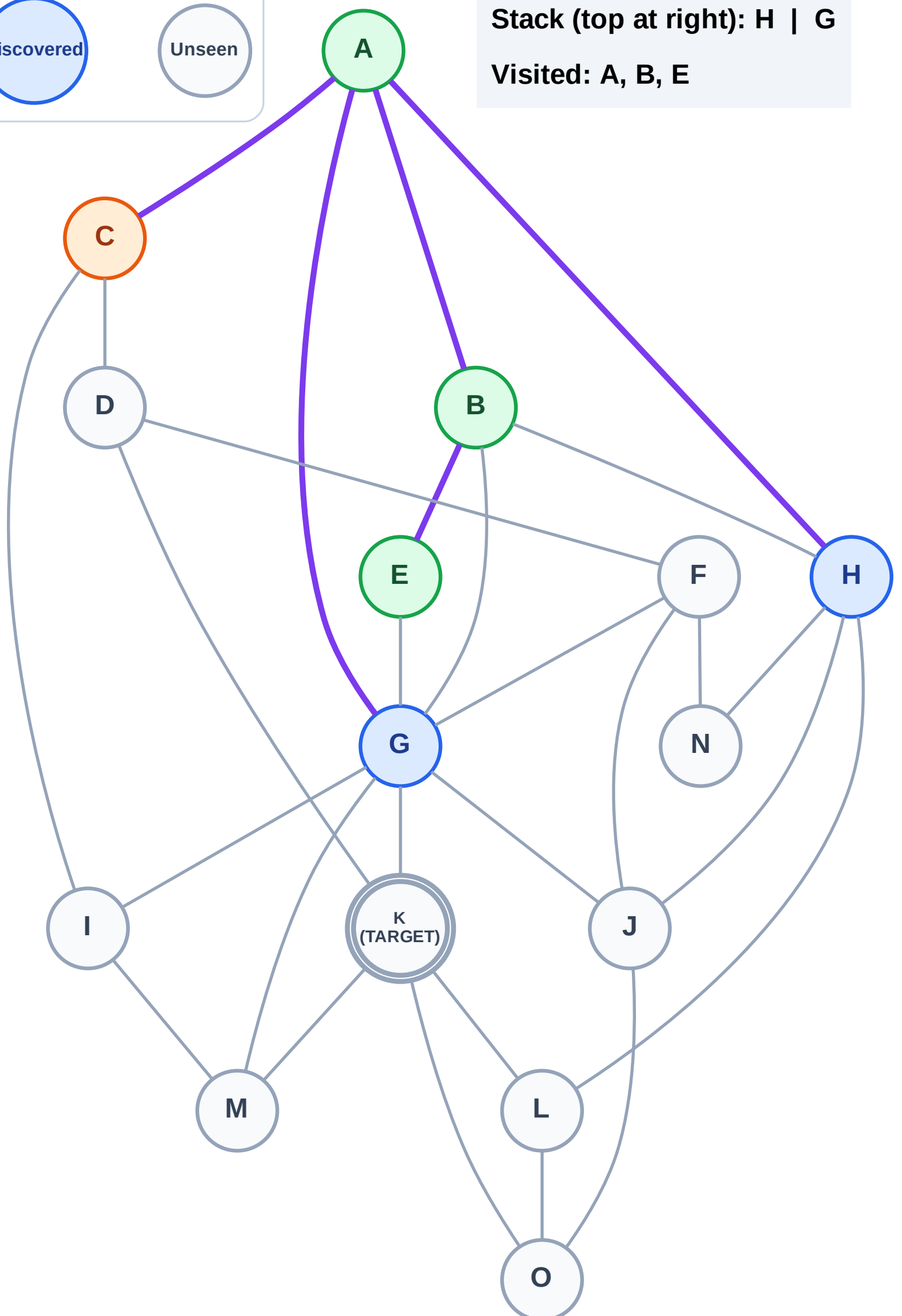


Step 8: Process node C

Legend

Unseen

Visited: A, B, E



DFS Traversal

Step 9: Discover I, D from C

Legend



Complete



Processing



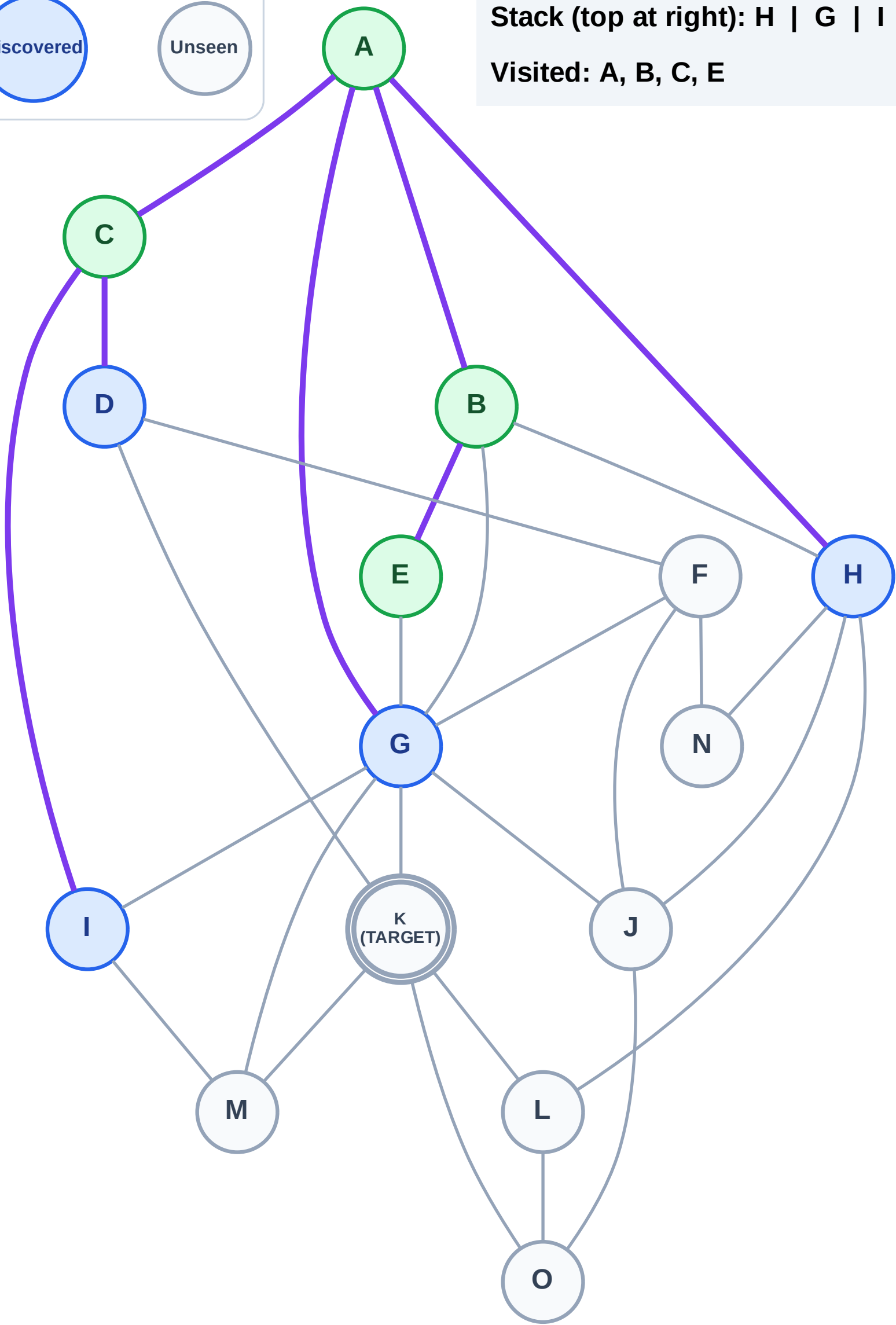
Discovered



Unseen

Stack (top at right): H | G | I | D

Visited: A, B, C, E



DFS Traversal

Step 10: Process node D

Legend

Complete

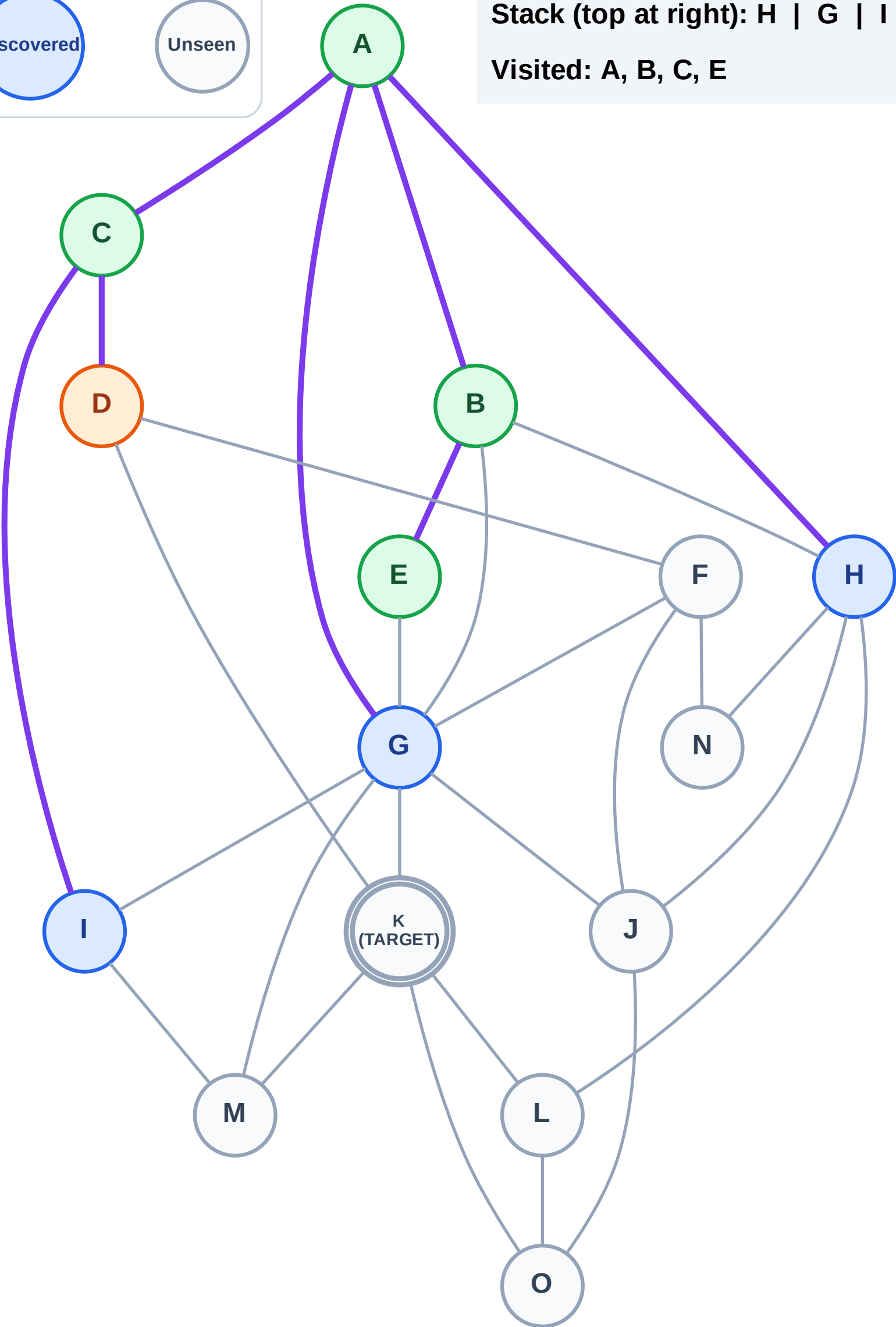
Processing

Discovered

Unseen

Stack (top at right): H | G | I

Visited: A, B, C, E



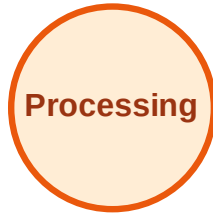
DFS Traversal

Step 11: Discover K, F from D

Legend



Complete



Processing



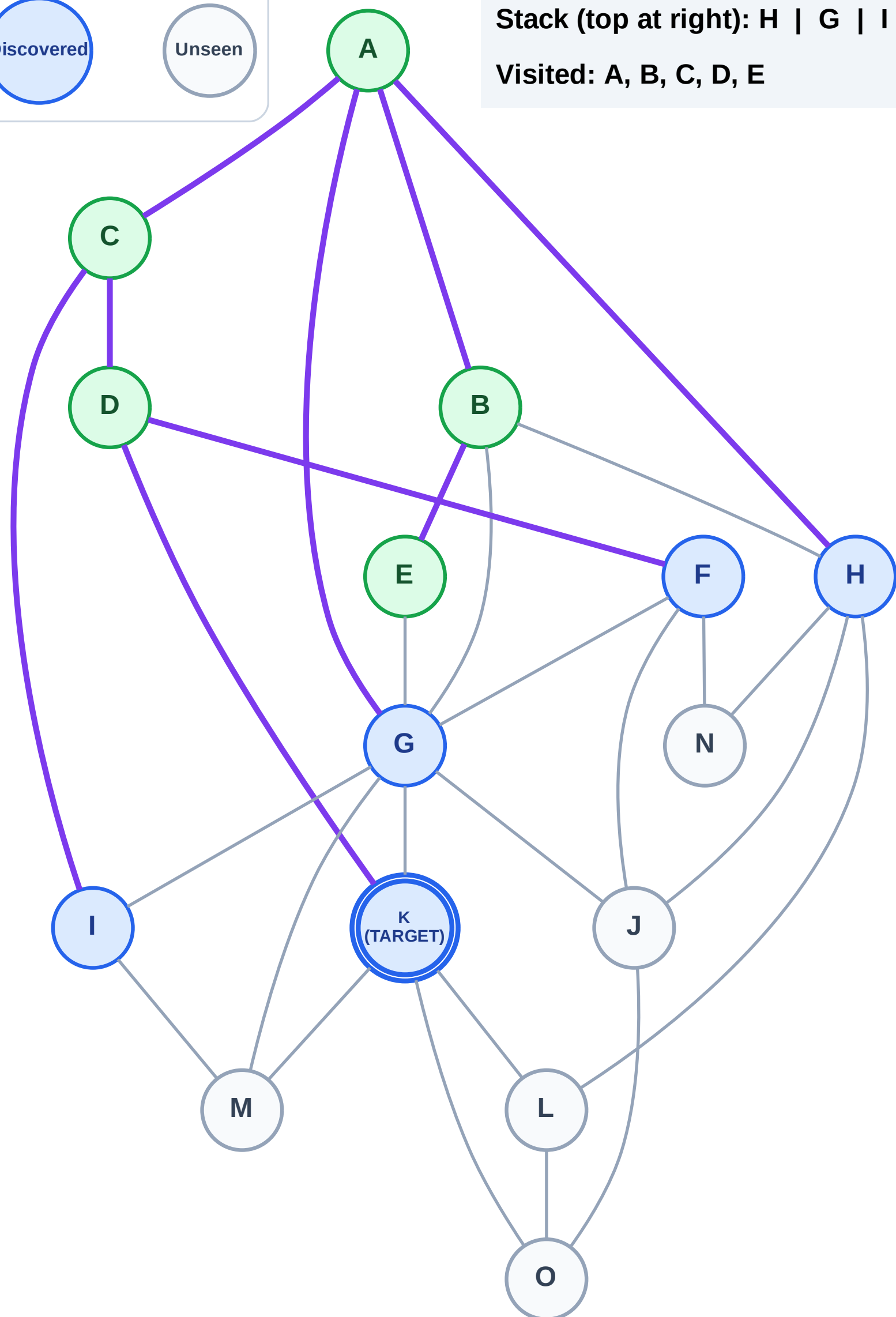
Discovered



Unseen

Stack (top at right): H | G | I | K | F

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node F

Legend

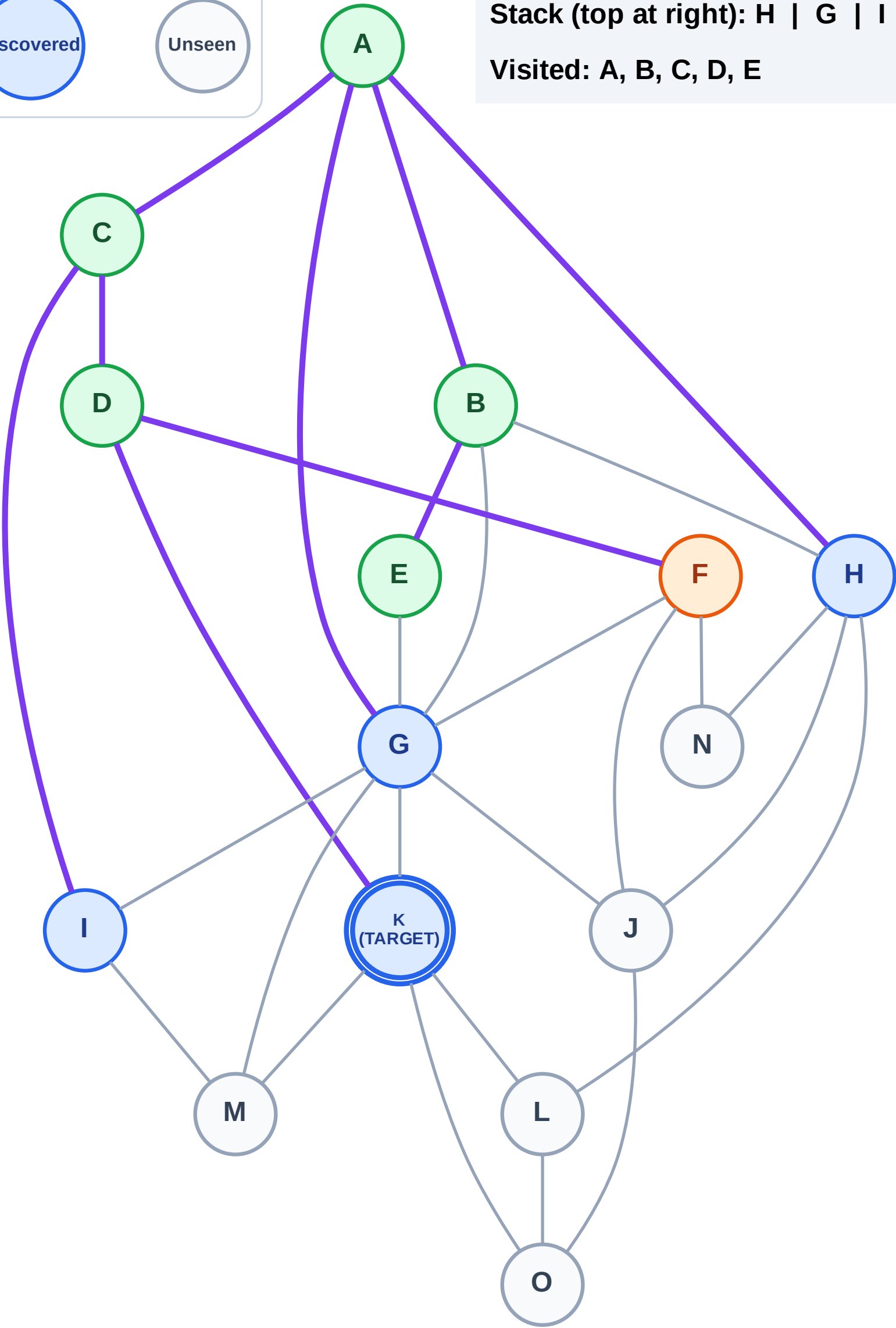
Complete

Processing

Discovered

Unseen

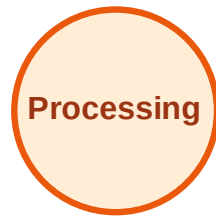
Stack (top at right): H | G | I | K
Visited: A, B, C, D, E



DFS Traversal

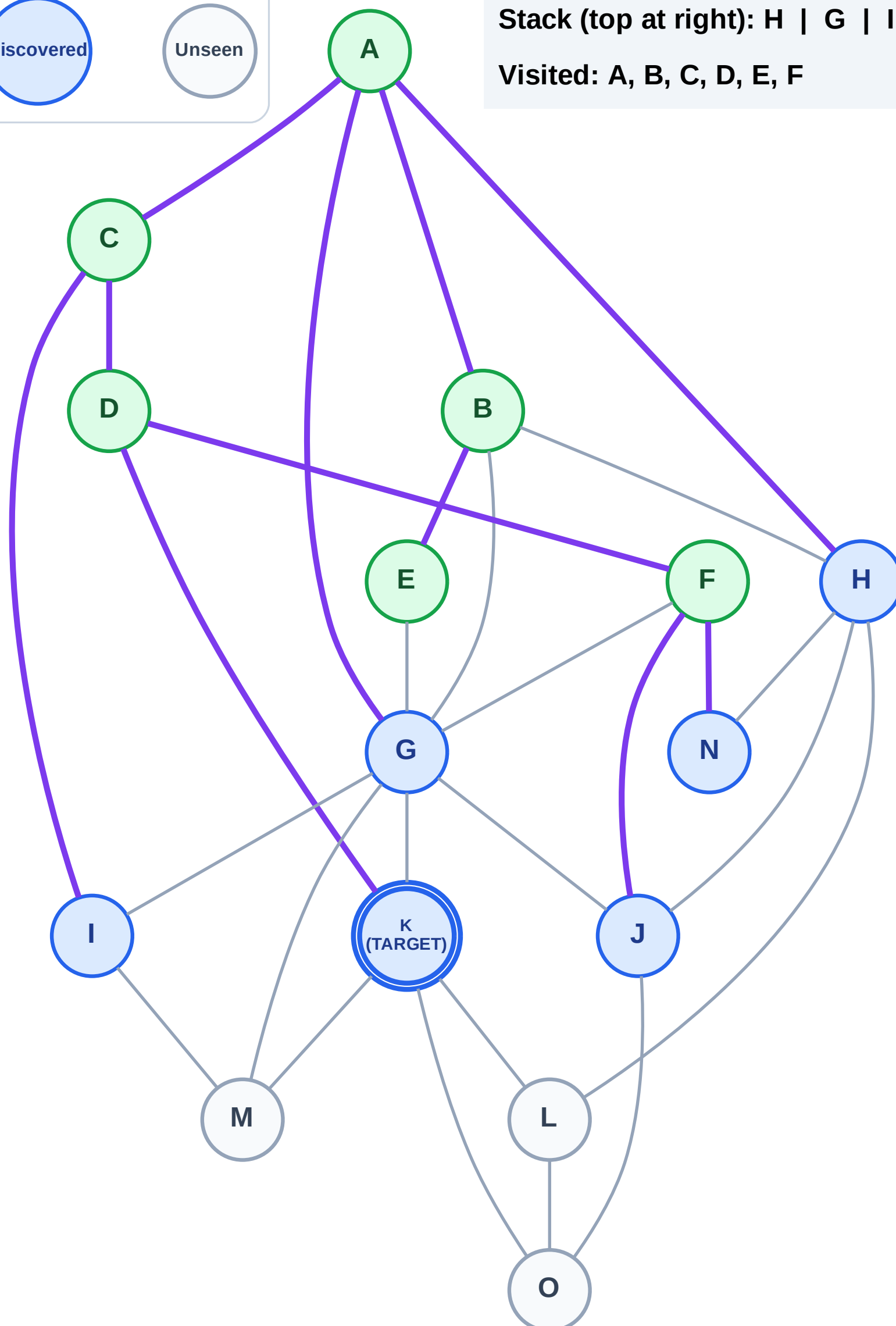
Step 13: Discover N, J from F

Legend



Stack (top at right): H | G | I | K | N | J

Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node J

Legend

Complete

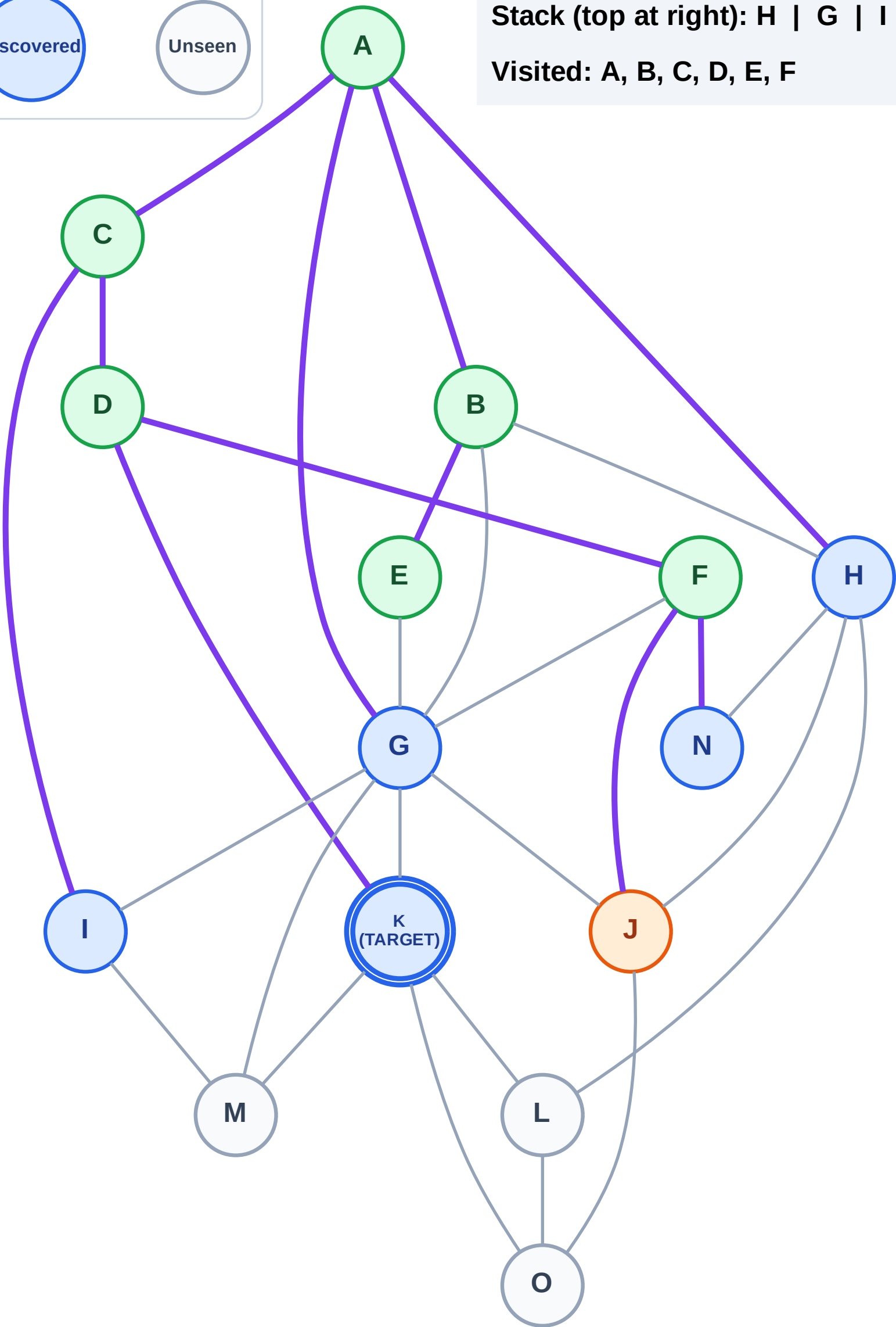
Processing

Discovered

Unseen

Stack (top at right): H | G | I | K | N

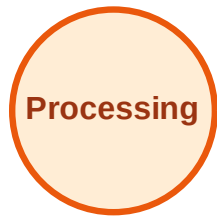
Visited: A, B, C, D, E, F



DFS Traversal

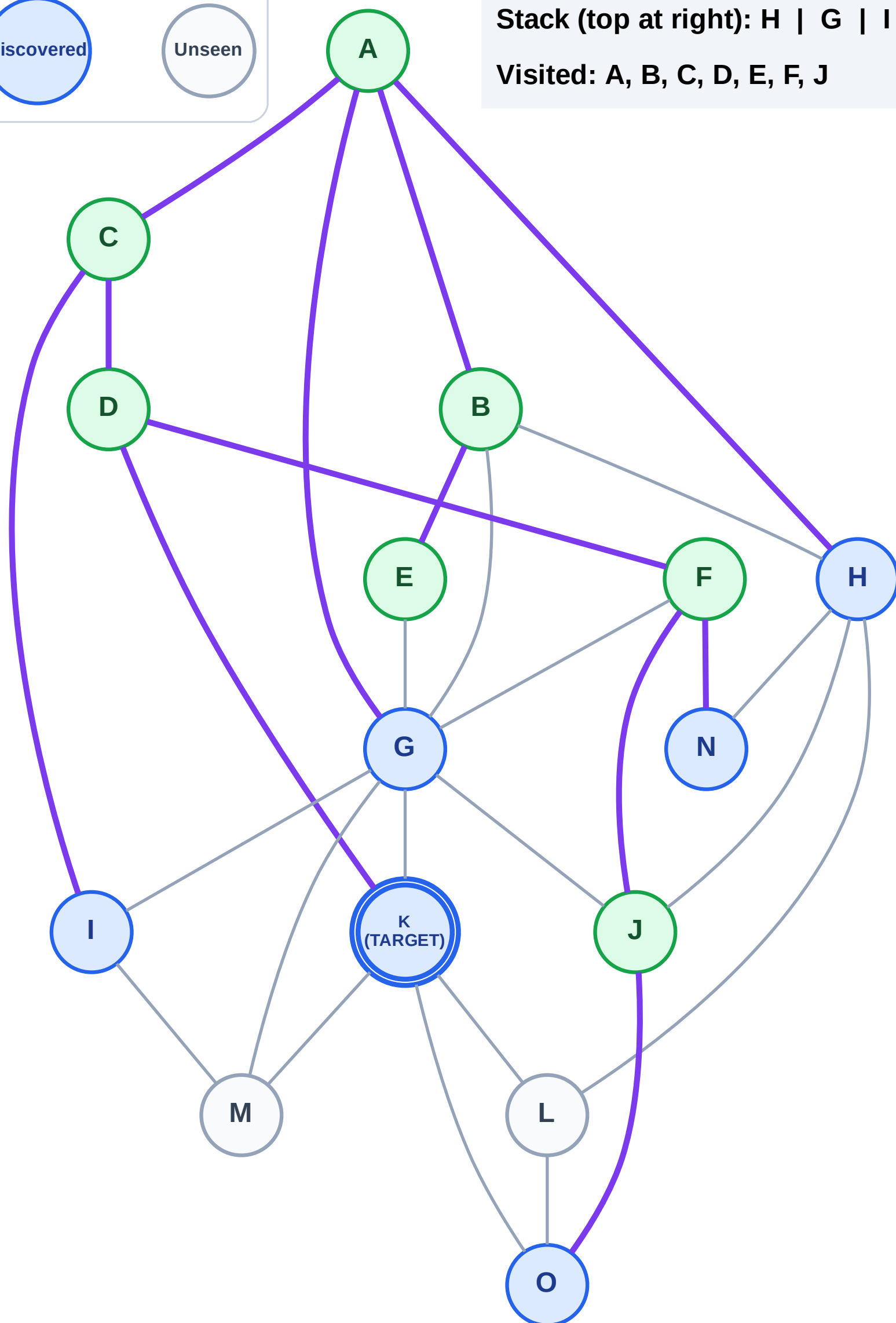
Step 15: Discover O from J

Legend



Stack (top at right): H | G | I | K | N | O

Visited: A, B, C, D, E, F, J



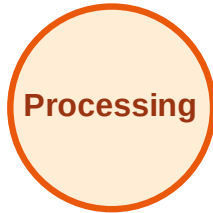
DFS Traversal

Step 16: Process node O

Legend



Complete



Processing



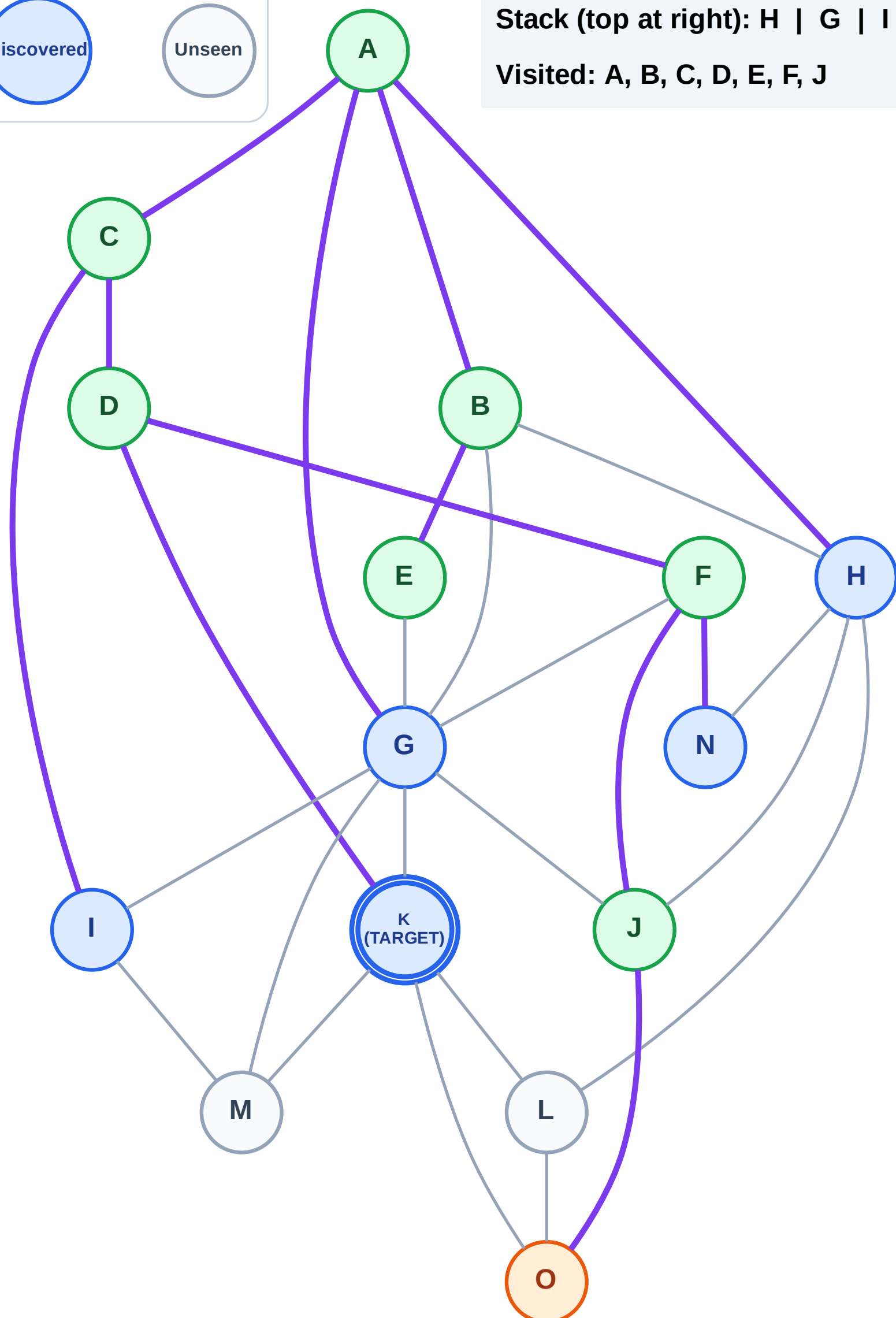
Discovered



Unseen

Stack (top at right): H | G | I | K | N

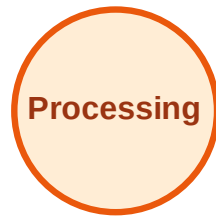
Visited: A, B, C, D, E, F, J



DFS Traversal

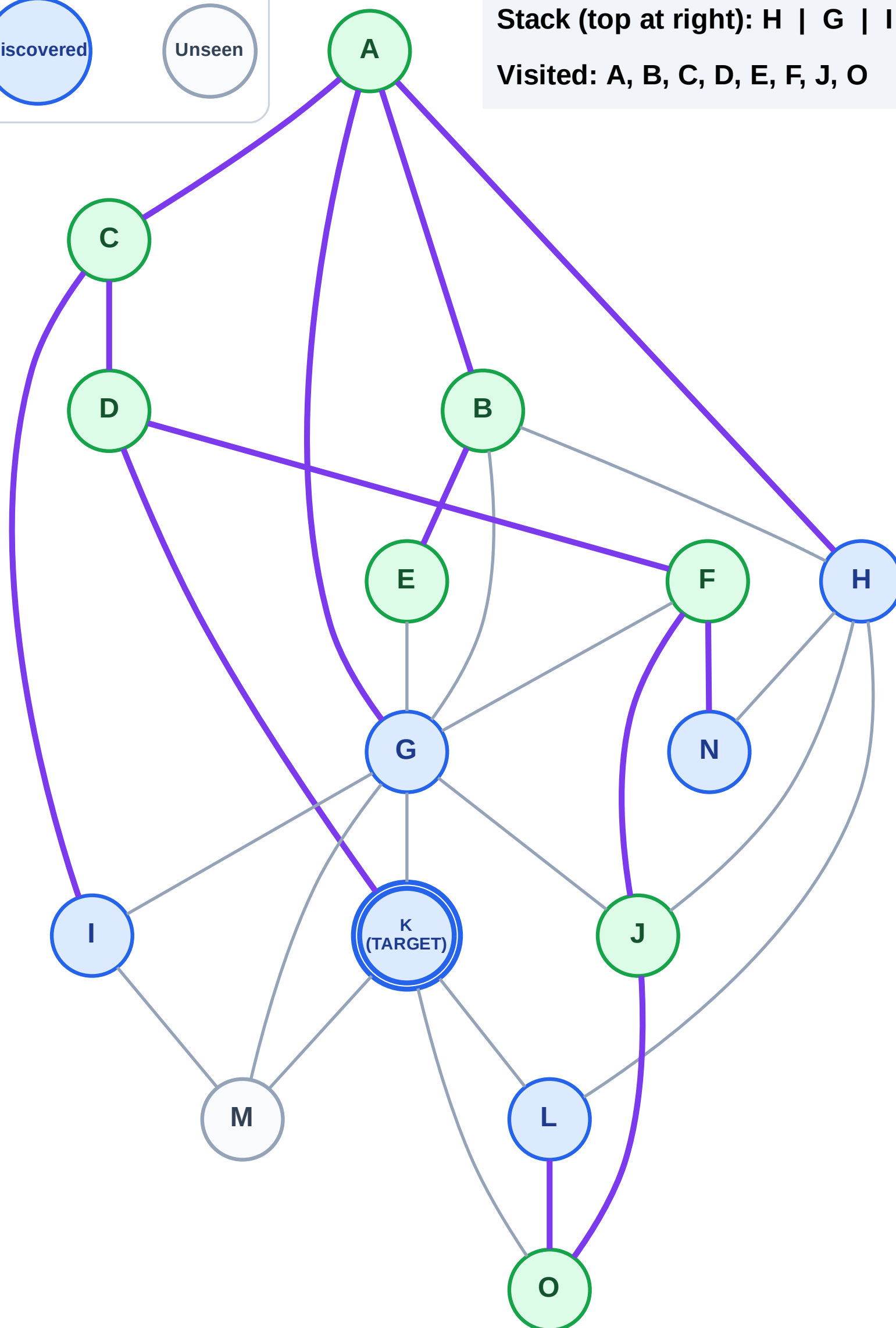
Step 17: Discover L from O

Legend



Stack (top at right): H | G | I | K | N | L

Visited: A, B, C, D, E, F, J, O



DFS Traversal

Step 18: Process node L

Legend

Complete

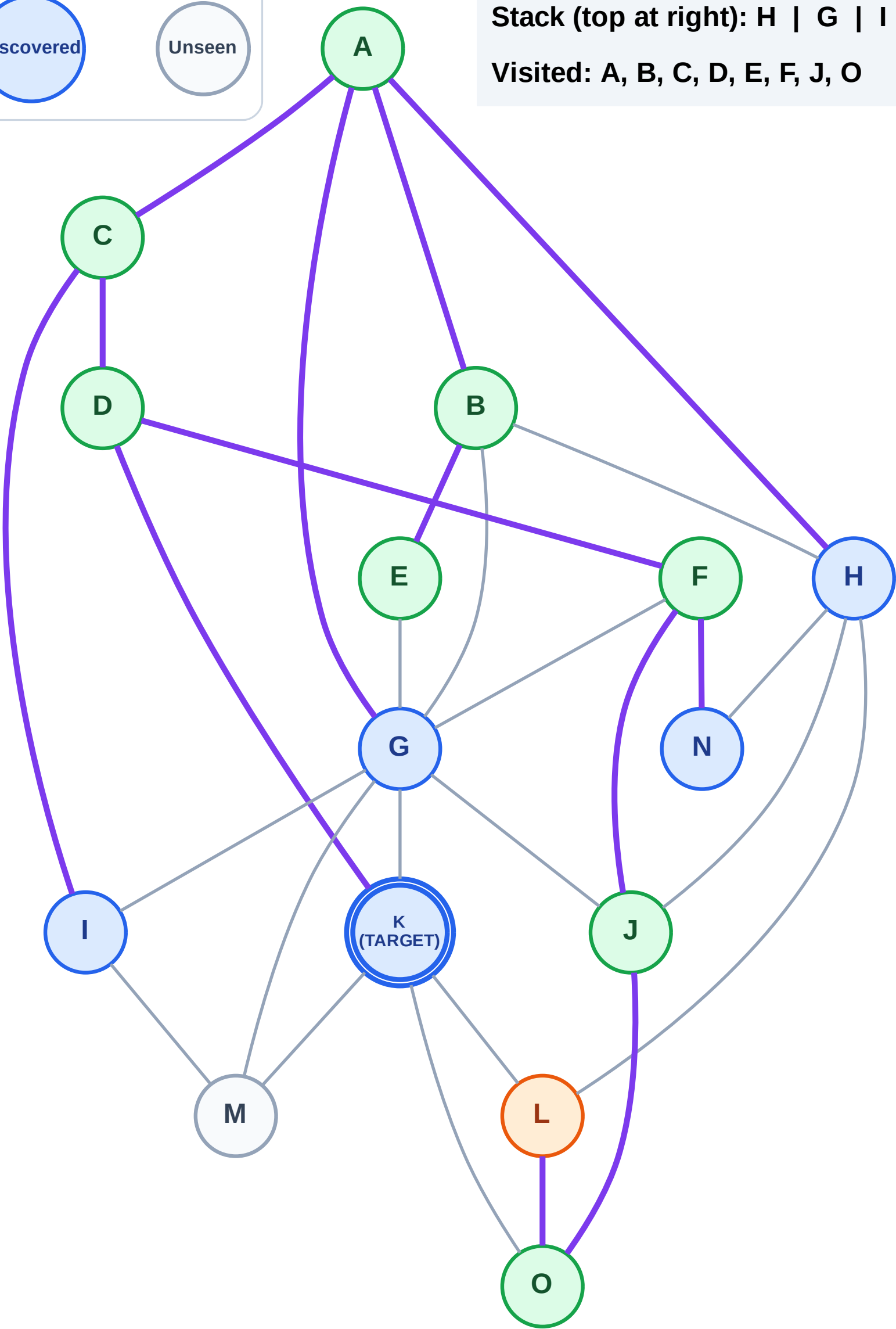
Processing

Discovered

Unseen

Stack (top at right): H | G | I | K | N

Visited: A, B, C, D, E, F, J, O



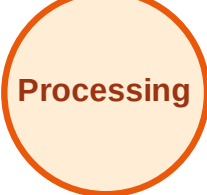
DFS Traversal

Step 19: No new neighbors from L

Legend



Complete



Processing



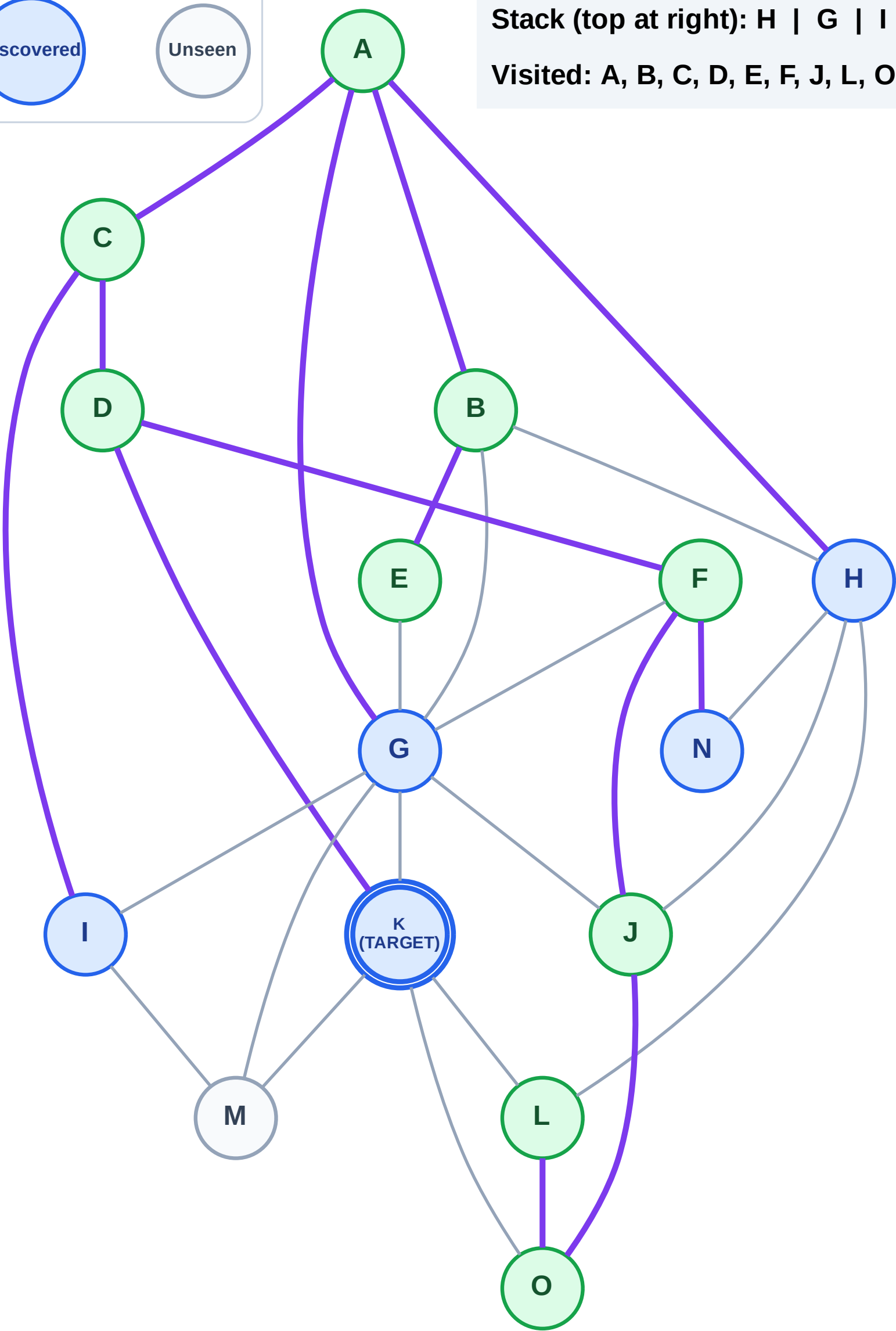
Discovered



Unseen

Stack (top at right): H | G | I | K | N

Visited: A, B, C, D, E, F, J, L, O



DFS Traversal

Step 20: Process node N

Legend



Complete



Processing



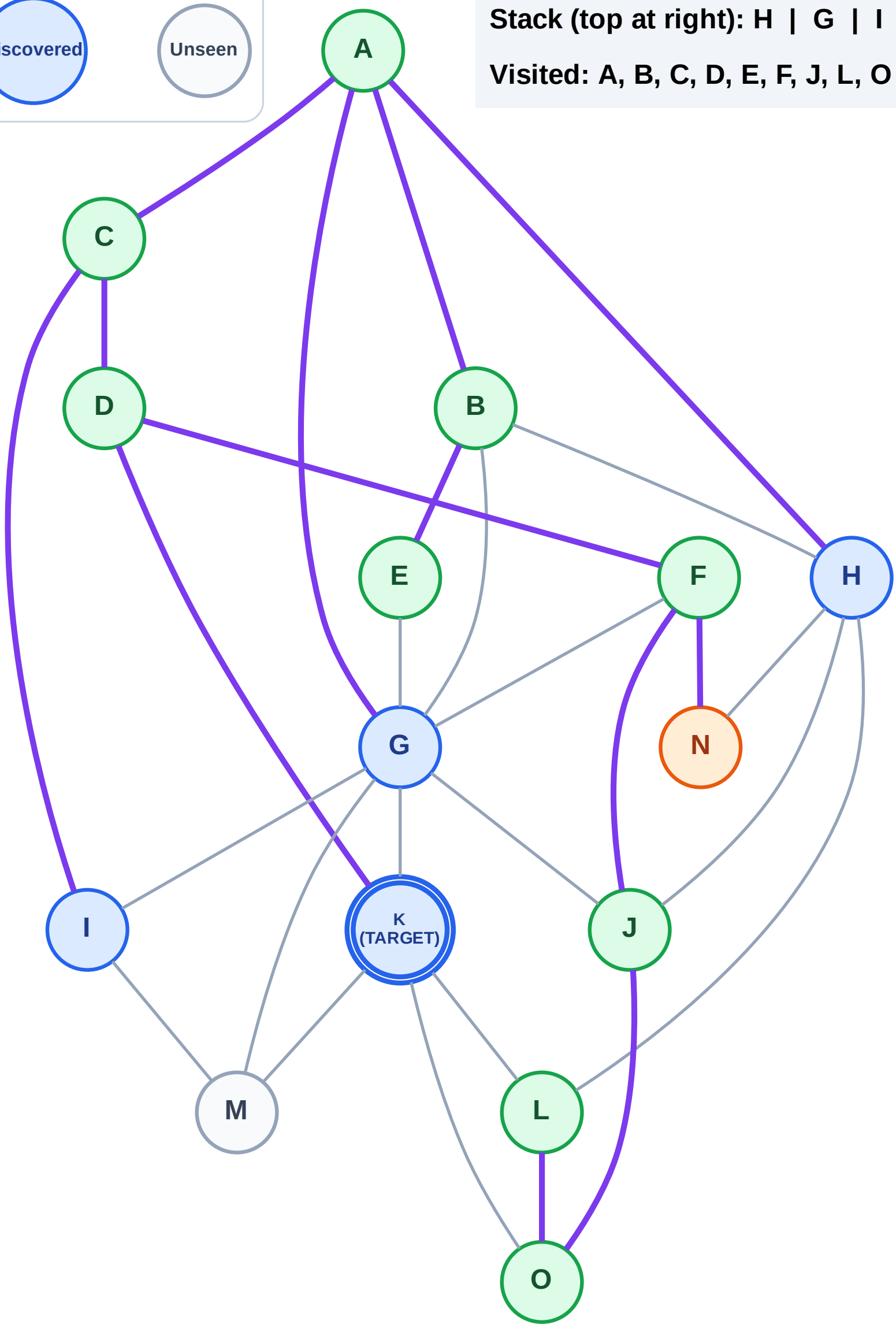
Discovered



Unseen

Stack (top at right): H | G | I | K

Visited: A, B, C, D, E, F, J, L, O



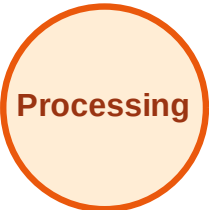
DFS Traversal

Step 21: No new neighbors from N

Legend



Complete



Processing



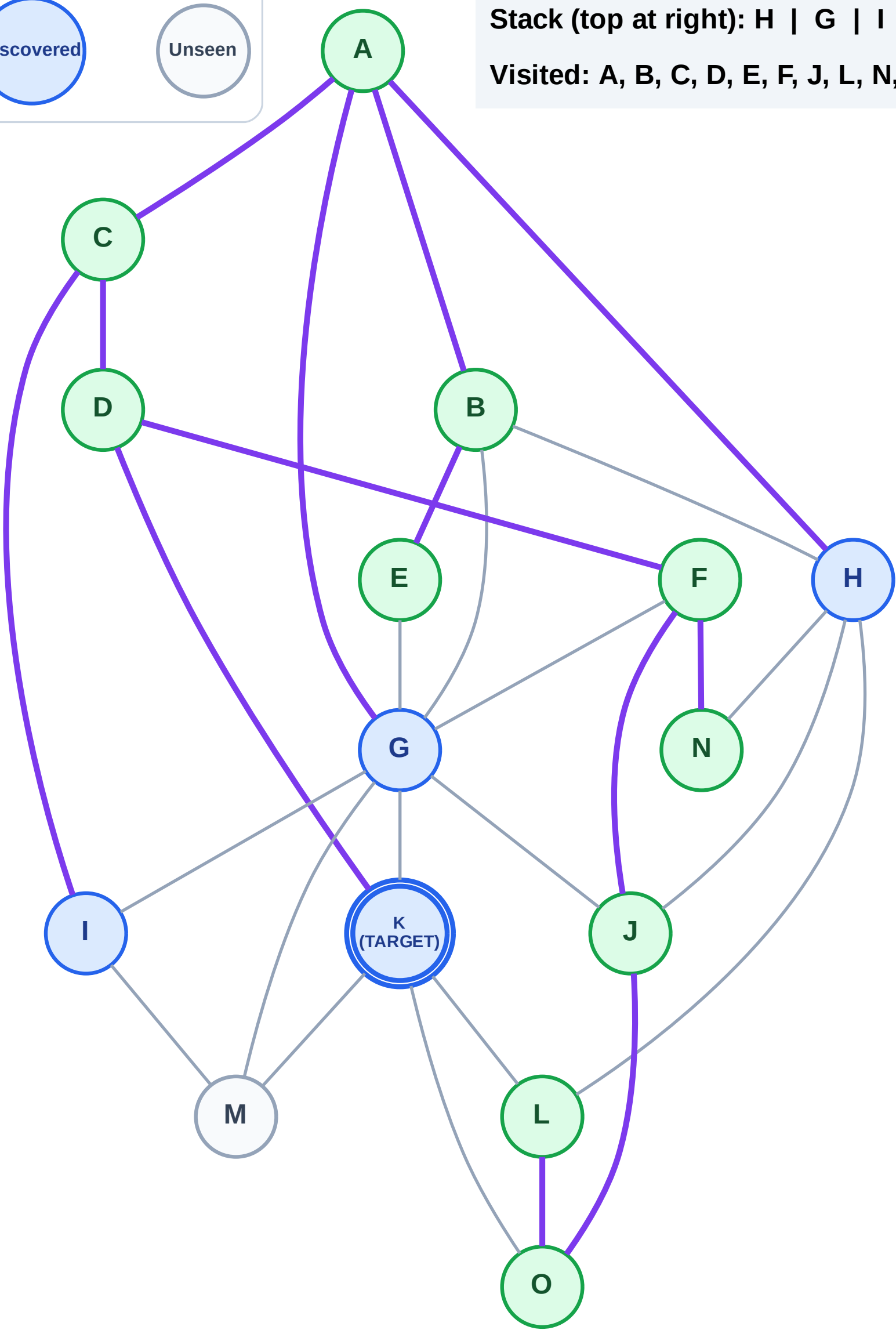
Discovered



Unseen

Stack (top at right): H | G | I | K

Visited: A, B, C, D, E, F, J, L, N, O



DFS Traversal

Step 22: Process node K

Legend

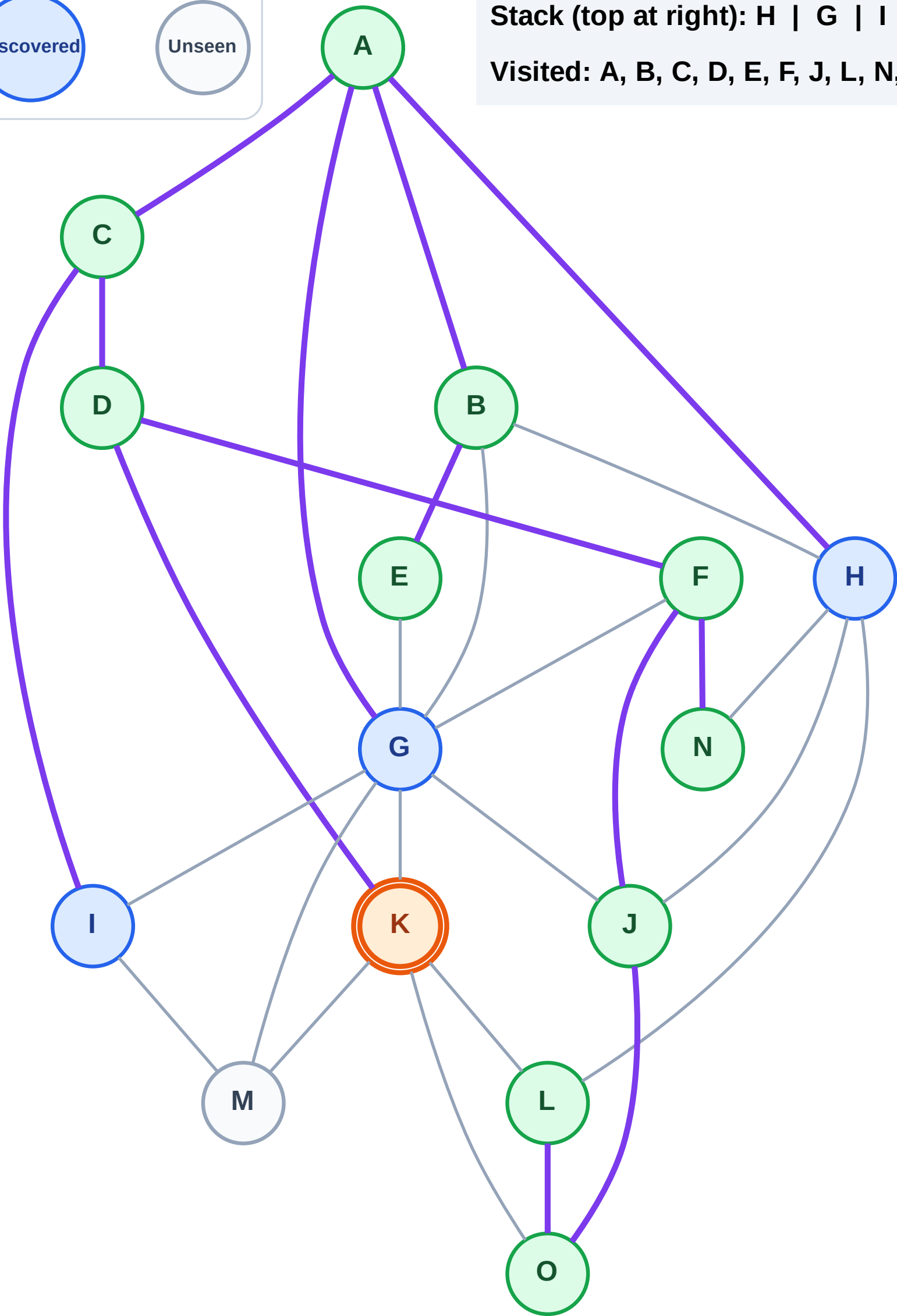
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | I
Visited: A, B, C, D, E, F, J, L, N, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (3 hops): A → C → D → K

Legend



Stack (top at right): H | G | I

Visited: A, B, C, D, E, F, J, K, L, N, O

